

Developing a Generative AI Model to Enhance Sentiment Analysis for the Saudi Dialect

by

Sulaiman Aftan, B.S., M.S.

A Dissertation

In

Computer Science

Submitted to the Graduate Faculty
of Texas Tech University in
Partial Fulfillment of
the Requirements for
the Degree of

DOCTOR OF PHILOSOPHY

Approved

Yu Zhuang, Ph.D.
Chair of the Committee

Yong Chen, Ph.D.

Victor Sheng, Ph.D.

Tara Salman, Ph.D.

Kulsawasd Jitkajornwanich, Ph.D.

Mark Sheridan, Ph.D.
Dean of the Graduate School

December, 2024

Copyright 2024, Sulaiman Aftan

ACKNOWLEDGMENTS

Firstly, I would like to express my deepest gratitude to Almighty Allah for granting me the health and strength to complete this Ph.D. dissertation.

I am also sincerely thankful to the Saudi Arabian Cultural Mission (SACM) in the U.S. for sponsoring my Ph.D. studies at Texas Tech University. Their financial support allowed me to focus entirely on my academic journey without any concerns.

I want to express my gratitude to my committee chair, Dr. Zhuang, for his unwavering guidance and support throughout this process. I am equally grateful to Dr. Chen, Dr. Sheng, Dr. Salman, and Dr. Jitkajornwanich for their academic contributions, which played a critical role in shaping my dissertation. Their constant feedback and insights were invaluable in directing my research.

I also want to express my profound gratitude to my family – my father and mother, my wife, my four children, and my brothers and sisters – who have always believed in me. Their support and encouragement have been a constant source of motivation in my pursuit of academic excellence.

Lastly, I would like to thank the Department of Computer Science within the Edward E. Whitacre Jr. College of Engineering, as well as the TTU Library, for providing access to journals, IT resources, and other essential tools that supported my research and the completion of my dissertation.

TABLE OF CONTENTS

ACKNOWLEDGMENTS	ii
ABSTRACT	v
LIST OF TABLES	vi
LIST OF FIGURES.....	vii
LIST OF ABBREVIATIONS.....	viii
1. INTRODUCTION	1
1.1 Research Motivation.....	1
1.2 Research Objectives.....	5
1.3 Structure of the Paper.....	6
2. LITERATURE REVIEW.....	7
2.1 Introduction.....	7
2.2 Artificial Intelligence and Natural Language Processing.....	8
2.3 The Nature of Dialects.....	9
2.4 Comparative Analysis of Saudi and Other Arabic Dialects.....	11
2.5 Role of AI in Studying the Saudi Dialect.....	14
2.6 Sentiment Analysis and Machine Learning Models.....	15
2.7 BERT Model.....	17
2.8 Sentiment Analysis of Saudi Dialect.....	20
2.9 Machine Learning Techniques in Saudi Dialect Analysis.....	23
2.10 Hybrid Approaches to Sentiment Analysis.....	24
3. METHODOLOGY.....	26
3.1 Introduction.....	26
3.2 The Architecture of the Saudi Dialect Sentiment Analysis Model.....	26
3.3 X Dataset.....	30
3.4 Annotation of the X Dataset.....	33
3.5 Application of GPT Model for Saudi Dialect Generation.....	36
3.6 GPT Generated Dataset.....	39

3.7 Use of AraBERT Model for Sentiment Analysis.....	42
4. MODEL ANALYSIS AND RESULTS.....	45
4.1 Introduction.....	45
4.2 MARBERT Model Performance.....	45
4.3 eXplainable AI (XAI) on MARBERT.....	47
4.4 AraGPT Model Performance.....	49
4.5 AraBERT Performance on Sentiment Analysis.....	51
5. CONCLUSIONS.....	57
5.1 Introduction.....	57
5.2 Main Findings and Implications.....	57
5.3 Limitations and Sentiment Analysis Challenges of SD.....	59
5.4 Opportunities in SD Analysis.....	63
5.5 Future Research.....	66
REFERENCES.....	67

ABSTRACT

Sentiment Analysis (SA) is a fundamental task in Natural Language Processing (NLP) with broad applications across various real-world domains. While Arabic is a globally significant language with several well-developed NLP models for its standard form, achieving high performance in sentiment analysis for the Saudi Dialect (SD) remains challenging. A key factor contributing to this difficulty is inadequate SD datasets for training of NLP models. This study introduces a novel method for adapting a high-resource language model to a closely related but low-resource dialect by combining moderate effort in SD data collection with generative AI to address this problem of inadequacy in SD datasets. Then, AraBERT was fine-tuned using a combination of collected SD data and additional SD data generated by GPT. The results demonstrate a significant improvement in SD sentiment analysis performance compared to the AraBERT model, which is fine-tuned with only collected SD datasets. This approach highlights an efficient approach to generating high-quality datasets for fine-tuning a model trained on a high-resource language to perform well in a low-resource dialect. Leveraging generative AI enables reduced effort in data collection, making our approach a promising avenue for future research in low-resource NLP tasks.

Keywords: Generative AI, Sentiment Analysis, Saudi Dialect, NLP

LIST OF TABLES

1	Dialectal Differences in Shapes & Phonemes of some Arabic Letters.....	14
2	Summary of SD Datasets.....	21
3	Description of the Saudi Twitter Dataset (STD).....	32
4	Examples of Preprocessing Steps on STD Corpus.....	34
5	Description of the Generated Dataset.....	39
6	Examples of the Generated Saudi Dialect Dataset.....	41
7	MARBERT Model's Evaluation Metrics.....	46
8	AraBERT Model's Classification Report (after 1 Iteration).....	52
9	Accuracy and F1-Scores (after 10 Iterations of Reshuffling).....	54

LIST OF FIGURES

1	Research Papers published in the Domain of NLP with SD.....	5
2	NLP Market Share by Industry Sector.....	9
3	BERT Input Representation.....	18
4	Various Types of BERT.....	19
5	The Architecture of the SD Sentiment Analysis Model.	29
6	Most Visited Social Media Platforms in Saudi Arabia.....	30
7	Architecture of AraGPT Model.....	38
8	Confusion Matrix of the Annotated Dataset.....	46
9	Predicted Label Distribution of the Annotated Dataset.....	47
10	LIME Explanation of the MARBERT Model Prediction.....	49
11	Confusion Matrix of AraBERT.....	53
12	Chart of Average and Standard Deviation Scores for Datasets.....	56
13	SD Opportunities by using NLP.....	65

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ANLP	Arabic Natural Language Processing
AOC	Arabic Online Commentary
ASA	Arabic Sentiment Analysis
BERT	Bidirectional Encoder Representations from Transformers
BLEU	Bilingual Evaluation Understudy
BLSTM	Bidirectional LSTM
BPE	Byte Pair Encoding
CLSTM	Convolutional LSTM
CNN	Convolutional Neural Networks
CST	Saudi Communications, Space, and Technology Commission
DL	Deep Learning
FFNN	Feed-Forward Neural Networks
GEA	Saudi General Entertainment Authority
GPT	Generative Pre-trained Transformer
HRL	High-Resource Language
LIME	Local Interpretable Model-agnostic Explanations
LRL	Low-Resource Language
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstrum Coefficients
ML	Machine Learning
MLM	Masked Language Model

MLP	Multi-Layer Perceptrons
MNB	Multinomial Naïve Bayes
MSA	Modern Standard Arabic
NB	Naive Bayes
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
POS	Part-Of-Speech
RNN	Recurrent Neural Networks
STD	Saudi Twitter Dataset.
SVM	Support Vector Machines
VGt	Verbal Guise Technique
XAI	Explainable AI

CHAPTER I

INTRODUCTION

1.1 Research Motivation

Global Language Landscape

As of October 2024, over seven thousand languages are spoken globally. Among them, English and Arabic rank as the most widely used. There are an estimated 1.5 billion speakers for English, and 310 million speakers for Arabic. This includes both native and non-native categories. Each language encompasses distinct properties, characteristics, and variations in accents or dialects [1], [2]. A dialect can be described as a linguistic phenomenon specific to a geographical area, evolving and being shared by its inhabitants. These linguistic properties include the pronunciation and articulation of letters, the positioning of the speech organs for specific sounds, the modulation of soft sounds, the use of inflections, and the interaction of adjacent sounds [3], [4]. When these characteristics are prevalent within a particular region, they form dialects. Each area possesses a unique dialect that distinguishes it from others.

Understanding Arabic Dialects

Arabic dialects refer to regional or social varieties of the Arabic language that exhibit differences in pronunciation, vocabulary, idioms, expressions, meanings, and grammatical structures when compared to the standard or dominant form of Arabic. The

development of dialects can be attributed to factors such as geographical isolation, historical contexts, cultural influences, and interactions with other languages. Among other global languages, English also holds significant importance in communication [5], [6]. Both English and Arabic possess rich historical and cultural backgrounds. However, there are significant differences in their structures, writing systems, dialects, and contexts of use.

A comprehensive understanding of these distinctions can substantially enrich one's appreciation of both languages and the cultures they embody [7]. Dialects play a vital role in interpersonal communication, community interactions, tribal dialogues, national discussions, consumer behavior, business transactions, and both social and antisocial networks [6], [8]. Business and social platforms serve as the primary arenas for dialect usage because they generate billions of datasets daily. This abundance complicates the task of distinguishing dialects from standard grammatical structures.

For non-native speakers, differentiating between standard and dialectal sentences in Arabic can be particularly challenging. Among the various computational, linguistic, and mathematical approaches, technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) offer automated detection capabilities. In particular, the primary phase of speech recognition is important because it involves classifying Arabic dialects and recognizing the accents of speakers in any spoken language [6], [9].

The Saudi Dialect in Context

The Arabic dialects include the Saudi Dialect (SD). These dialects are among the most widely spoken varieties of Arabic. Regarding SD, there are over 18 million speakers globally. They are primarily concentrated in Saudi Arabia and the broader Middle East. Understanding the intricacies of the SD can be challenging in diverse contexts such as events, activities, behaviors, and other domains. ML techniques have been effectively applied in classifying these dialects, achieving varying degrees of accuracy. For example, an analysis of 32,063 tweets in Arabic dialects using Long Short-Term Memory (LSTM), Bidirectional LSTM (BLSTM), and Support Vector Machines (SVM) revealed that LSTM and BLSTM outperformed SVM, achieving accuracies of 94%, 92%, and 86.4%, respectively [10]. However, traditional ML methods often struggle to accurately distinguish between dialectal and non-dialectal sentences due to the complexities associated with vast datasets, deep semantic meanings, morphological intricacies, and other linguistic challenges.

Advances in Deep Learning for Arabic NLP

Researchers have explored various DL methods to address NLP tasks within the Arabic language. Models such as AraBERT and MARBERT are built upon Bidirectional Encoder Representations from Transformers (BERT). AraBERT and MARBERT have demonstrated significant success in processing standard Arabic [11], [12], [13]. However, preliminary applications of AraBERT for sentiment analysis in the Saudi dialect yielded

unsatisfactory results. Hence, the effectiveness of various Arabic NLP models in handling dialects remains uncertain.

Gaps in Research

According to Figure 1, the number of research articles dedicated to Arabic dialects (SD in particular) is low, with only 110 research articles published using NLP techniques and approaches. Only seven papers were published between 2009 and 2023. This is a significantly smaller number compared to those published in other English and French dialects [14]. This indicates a substantial gap in the literature concerning the research and study of Arabic dialects, especially within Saudi Arabia. Further extensive research employing NLP technologies and methodologies is necessary to deepen the understanding of the complex structure of the SD. This includes both machine learning and deep learning. Continued investigation in this domain has the potential to greatly advance linguistic research and enhance language processing systems. Collaboration between academics and institutions is vital to bridging this gap and contributing meaningful insights to the evolving field of Arabic dialect studies.



Figure 1: Research Papers published in the Domain of NLP with SD

1.2 Research Objectives

There is a significant deficiency in research concerning the annotation and sentiment analysis of the Saudi Dialect (SD). As a result, there is a gap in data availability. To address this issue, a proposal is made to leverage generative AI to create a new dataset utilizing the GPT model. The objective is to enhance the capabilities of deep learning models like AraBERT in executing complex tasks such as sentiment analysis.

Methodological Proposal for Fine-Tuning AraBERT

To improve sentiment analysis outcomes in the Saudi dialect, the goal is to fine-tune the AraBERT model using datasets specific to the Saudi dialect. However, the process of collecting a substantial amount of high-quality SD data suitable for NLP processing can be time-consuming and labor-intensive. Therefore, a strategy is proposed that combines moderate efforts in collecting SD data with the application of generative AI to create synthetic SD data. This combined dataset can be utilized to fine-tune AraBERT for sentiment analysis in the Saudi dialect.

1.3 Structure of the Paper

The remainder of this paper is structured as follows: Chapter II provides a literature review summary regarding language dialects, the role of AI and NLP in studying SD, sentiment analysis and machine learning models related to SD. Chapter III contains the step-by-step methodological approach to generate the SD dataset. Chapter IV provides a deeper look into the model performances, results and evaluation scores. Finally, Chapter V concludes with a discussion of the main findings, limitations, challenges, opportunities in SD analysis, and future recommendations.

CHAPTER II

LITERATURE REVIEW

2.1 Introduction

The identification of the Saudi Dialect (SD) is receiving considerable attention within the realms of deep learning (DL) and Natural Language Processing (NLP). This heightened interest is largely due to the increasing prevalence of dialectal Saudi Arabic text used for both formal and informal communication across various platforms – including online interactions, events, and business activities. Consequently, there is a pressing need for research focused on dialectal corpora, as well as language classification and identification methodologies.

The primary dialects prevalent in Saudi Arabia are Hijazi, Najdi, Janobi, Hasawi, and Southern dialects. The Hijazi dialect is prominent in the western region. The Najdi dialect originated from the central region. The Janobi and the Hasawi dialects are also associated with central Saudi Arabia. In addition, the Gulf Arabic dialect is spoken in the eastern part of the country, while southern dialects are common in the southern region. These dialects share the same Arabic characters as those used in neighboring countries such as Yemen, Egypt, the UAE, Qatar, and Bahrain, yet they often do not adhere to the grammatical structures characteristic of Modern Standard Arabic (MSA) [15]. In response to these challenges, various researchers have implemented advanced technologies. This includes DL models to address tasks such as generating, classifying,

and recognizing the SD. ML models have been employed to classify dialects based on extensive datasets, achieving varying accuracy levels [10], [16]. Moreover, research efforts have utilized DL techniques for Arabic dialect annotation and sentiment analysis.

2.2 Artificial Intelligence and Natural Language Processing

Artificial Intelligence (AI) possesses the capability to analyze data, recognize patterns, and develop predictive models. These functions are essential for creating sustainable behaviors within the circular economy framework [17]. Natural Language Processing (NLP) is a prominent branch of AI. NLP focuses on how computers can comprehend and interact with human language. This enables efficient and meaningful analysis, comprehension, and generation of human language [18]. NLP has diverse applications across various industries. These include sentiment analysis, language translation, chatbots, virtual assistants, information extraction, and text summarization.

The distribution of NLP's market share spans multiple industry sectors. Significant proportions are in business and legal services. They account for 26.5% of the total market share. This indicates a strong reliance on NLP technologies within these fields. The media and entertainment industry also holds a considerable share at 21.2% showing the importance of NLP in content creation and delivery. The energy sector represents 15.01% of the NLP market, while the healthcare and finance industries follow with 8.86% and 8.25%, respectively. This demonstrates their utilization of NLP for managing healthcare data and conducting financial analyses. Other sectors of security,

transportation, and retail, represent smaller proportions of 5.82%, 4.39%, and 4.12%, respectively, with the remaining sectors collectively accounting for 5.86% of the NLP market [19].

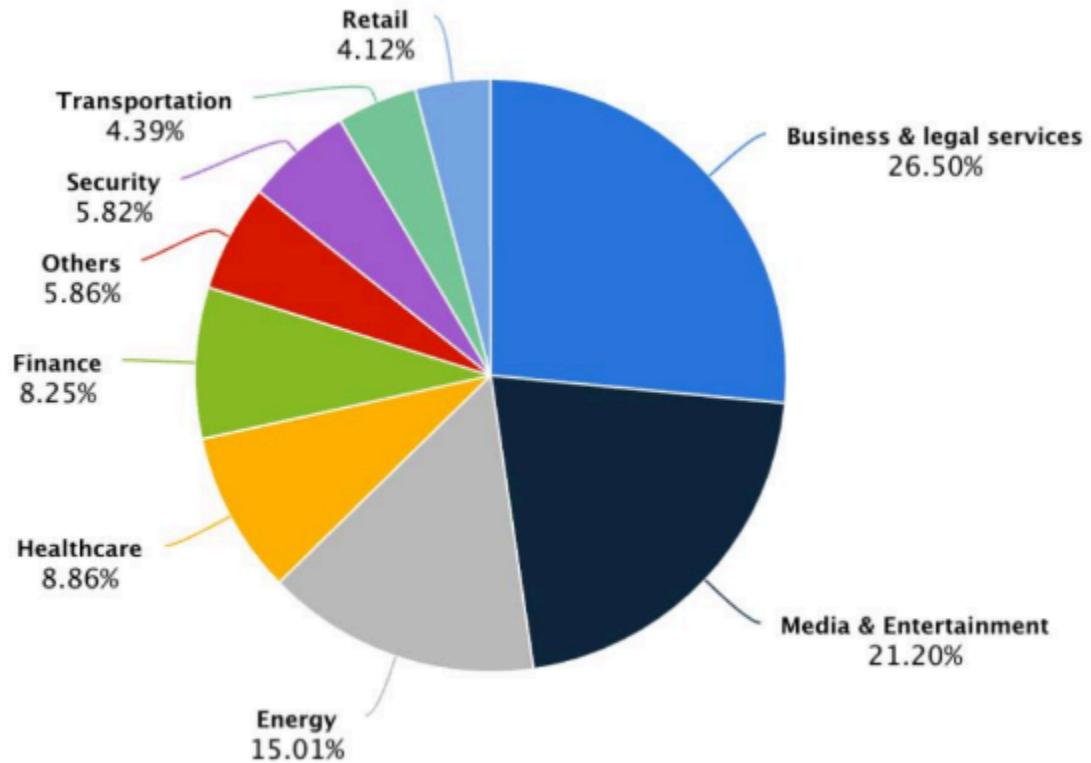


Figure 2: NLP Market Share by Industry Sector

2.3 The Nature of Dialects

A dialect is defined as a distinct form of a language specific to a particular population of speakers or a geographical region. Dialects are characterized by variations in vocabulary, grammar, and cultural nuances [18]. These linguistic variations often

emerge from historical, social, and geographical contexts. They are influenced by events such as migrations and interactions among different linguistic communities [20]. Social factors such as socioeconomic background, the speech of linguistic scholars, and cultural influences, contribute to the formation of dialects. Regional variations in the speech of leaders and elders may also give rise to distinct dialects among specific groups of speakers [21].

Arabic Dialect Research

Research into Arabic dialects has gained considerable traction over the past decade. Arabic dialects encompass a broad spectrum of linguistic variations spoken across the Arab world. They can be categorized into three primary groups: Maghrebi, Levantine, and Gulf dialects [21]. Maghrebi dialects are prevalent in North African countries, including Morocco, Algeria, and Libya [22]. Levantine dialects are found in the Eastern Mediterranean region, covering Lebanon, Syria, Jordan, and Palestine [23]. Gulf dialects are primarily spoken in the Arabian Gulf region. This region includes countries such as Saudi Arabia and the United Arab Emirates [24]. Each dialect possesses unique phonological, lexical, and grammatical characteristics, contributing to the rich diversity of the Arabic language. Although Modern Standard Arabic (MSA) serves as a unified written medium, everyday conversations often incorporate various distinct dialects [25].

Saudi Dialect Characteristics

The Saudi Dialect (SD) is a subset of Gulf Arabic dialects. It is predominantly spoken in the Kingdom of Saudi Arabia (KSA) [24]. It exhibits distinctive linguistic features that differentiate it from other Arabic variations. The SD is shaped by a blend of tribal, historical, and cultural influences with deep roots in the Arabian Peninsula [26]. Its unique character is further defined by phonological distinctions, lexical choices, and idiomatic expressions [26]. The evolution of the SD has also been influenced by societal issues such as the impact of Bedouin traditions and rapid modernization [27]. This fusion of old and new language characteristics is reflected in specific pronunciation differences, such as the pronunciation of the letter “qaf” – which is pronounced as a “q” in MSA but as a “g” in the SD. Understanding the complexities of the SD is vital for effectively implementing NLP methods and for non-native speakers seeking to grasp local cultural contexts [28].

2.4 Comparative Analysis of Saudi and Other Arabic Dialects

Complexity in Analyzing Arabic Dialects

Analyzing Arabic dialects across different regions is a complex task due to the diverse linguistic characteristics and historical influences present in each area. A thorough examination of the unique features and deviations in each dialect is essential for gaining a comprehensive understanding of the intricacies of the Arabic language. Dialects

can be shaped by various factors. These include historical interactions with other cultures, and resulting in distinct linguistic elements that reflect the region's cultural history.

For example, the Iraqi dialect is characterized by the incorporation of Persian and Turkish loanwords. It is coupled with unique pronunciation patterns for specific letters [29]. The Levantine dialect spans regions like Lebanon, Jordan, and Syria, and has developed a distinct lexicon and pronunciation style due to significant influences from French and Aramaic languages [30]. In North Africa, the Moroccan dialect is heavily influenced by Berber culture, and exhibits distinctive phonetic features and syntactic structures [31]. These linguistic influences highlight how historical and cultural exchanges have shaped each dialect's development.

Distinctive Features of the Saudi Dialect

The Saudi dialect (SD) stands out for its use of the classical Arabic lexicon and its preservation of antiquated linguistic traits. Unlike other dialects that have absorbed a multitude of foreign influences, the SD retains much of its classical vocabulary. It reflects a linguistic continuity that maintains connections to older forms of Arabic. This preservation of linguistic elements sets the SD apart from other regional dialects, making it a key subject for comparative studies aimed at understanding Arabic language diversity [32].

Comparative Analysis of Saudi, Lebanese, and Moroccan Dialects

Comparing and contrasting dialects such as the Saudi, Moroccan, and Lebanese variants reveals distinct influences and linguistic characteristics. For example, the Lebanese dialect diverges from the SD mainly due to the presence of French and Aramaic influences, which have shaped its vocabulary and pronunciation. Meanwhile, the Moroccan dialect reflects significant Berber influences, making it phonetically and syntactically different from the SD. Through the comparative study of these dialects, scholars can better understand the broad spectrum of linguistic diversity within the Arabic language across different geographical areas [33-37].

Insights from Dialect Comparison

By examining the variations among these dialects, researchers can gain insights into the evolution and adaptation of the Arabic language over time and across regions. Table 1 illustrates the distinctive forms and phonemes of several Arabic characters across various dialects. It highlights the differences that arise from the interplay of historical, cultural, and linguistic influences. Understanding these differences provides a richer comprehension of the Arabic language's evolution, and how distinct regions have maintained or adapted classical elements in their speech.

Table 1 - Dialectal Differences in Shapes & Phonemes of some Arabic Letters

		Dialects			
		Iraqi	Levantine	Moroccan	Saudi
phonemes	/g/	گ	ج	ك	ج
	/tʃ/	چ	تس	تش	تس
	/v/	ف	ف	ف	ف
	/p/	ب	ب	ب	ب

2.5 Role of AI in Studying the Saudi Dialect

Utilizing AI for the study of the SD offers numerous benefits and opportunities to enhance linguistic analysis's effectiveness and comprehensiveness. Deep learning (DL) and NLP methods empower computers to analyze and understand human language. The SD can be analyzed with greater precision and efficiency. Recent studies have yielded promising results using NLP techniques to extract valuable insights from written and spoken texts in the SD. Given the SD's susceptibility to personal opinions and its inherent inconsistencies, this approach facilitates a deeper understanding that can significantly improve language analysis tasks related to the SD. For example, it enables accurate analysis of emotions expressed in social media posts, precise transcription of spoken conversations in the SD into written form, and effective labeling of SD text collections, thereby reducing the need for human intervention.

2.6 Sentiment Analysis and Machine Learning Models

In exploring sentiment analysis, various models have demonstrated efficacy in dialect classification. For instance, the Multi-Dialect Arabic Sentiment Twitter Dataset employed Support Vector Machines (SVM) and Long Short-Term Memory (LSTM) networks with notable success. This dataset was derived from Twitter interactions in the UAE and Egypt [8], [9]. The highest accuracy achieved for automatic dialect classification of the widely recognized Arabic Online Commentary (AOC) dataset reached 70% for the Egyptian dialect and 64.8% for the UAE dialect. This dataset includes multiple categories such as Egyptian, Gulf, Iraqi, Levantine, Maghrebi, and MSA, among others. Various ML classifiers such as LSTM, Convolutional Neural Networks (CNN), Bidirectional LSTM (BLSTM), and Convolutional LSTM (CLSTM), were used for classification. LSTM achieved a maximum accuracy of 71.4% [9]. In 2020, Alahmary et al. implemented DL techniques to create a sentiment analysis model for the SD using CNNs, which outperformed traditional ML methods such as SVM and Naive Bayes (NB) [38].

Contributions of AraCust and AraBERT

Further advancements in sentiment analysis have been made by Almuqren, who utilized AraCust to analyze 20,000 tweets. This study provides comprehensive insights into the corpus's construction, preprocessing, annotation processes, and features. Both AraCust and AraBERT demonstrated superior performance when compared to conventional NLP models, successfully predicting customer satisfaction for telecom

companies based on an analysis of Twitter data from 41.63 million subscribers [39], [40]. This study introduces a valuable dataset specifically designed for sentiment analysis within the Saudi Dialect highlighting its potential for future deep learning applications. The dataset is curated from tweets, and has proven instrumental in training models [16].

Challenges and the Role of Explainable AI

Despite these advancements, existing models often suffer from limitations related to the absence of linguistic domain expertise, which can hinder their interpretability and effectiveness in decision-making processes. To address these challenges, the emerging field of Explainable AI (XAI) is being utilized to produce outputs that incorporate insights from linguistic experts, thereby visualizing results for informed decision-making in various tasks. This includes classification and prediction [41]. However, there remains a shortage of research at the intersection of XAI and the Saudi Dialect, primarily due to the lack of newly generated SD datasets suitable for these models. Notably, one study [42] introduced an explainable framework for Arabic sentiment classification by integrating a Gaussian noise layer into DL models such as BiLSTM and CNN-BiLSTM, which mitigates overfitting and enhances performance. In addition, the study presents the Local Interpretable Model-agnostic Explanations (LIME) approach for Arabic Sentiment Analysis (ASA), providing accessible explanations for sentiment predictions. Another research effort [43] applied LIME to predict sentiment polarity in Arabic texts related to LASIK surgery, achieving an accuracy of 79.1% on the proposed dataset and validating the results by comparing word counts with probability weights in the context of ASA.

2.7 BERT Model

BERT (Bidirectional Encoder Representations from Transformers) is a deep learning model that processes text bidirectionally. It analyzes context from both the left and right sides simultaneously, unlike traditional models that process text unidirectionally. BERT was introduced by the Google AI language team in 2018, and is based on the Transformer architecture which employs multiple encoding layers. This model has demonstrated significant advantages in various Natural Language Processing (NLP) tasks, such as semantic understanding, inference, and text classification [45], [46]. The performance of BERT relies heavily on its Encoder, which includes mechanisms such as self-attention and feed-forward neural networks (FFNNs), while the Decoder contributes to transferring information across layers.

By utilizing the Transformer architecture, BERT has garnered widespread attention from researchers, leading to its application in numerous problems. BERT has consistently achieved state-of-the-art results compared to other computational models. BERT is a pre-trained model, which can be applied to various tasks using two key approaches: feature-based and fine-tuning. Pre-training involves using large-scale, unlabeled datasets, while fine-tuning refines the model's parameters using labeled data specific to downstream tasks. Notably, the model undergoes two primary phases: pre-training and fine-tuning. In the pre-training phase, BERT learns from unlabeled data using specific tasks relevant to the problem. During fine-tuning, the pre-trained model's parameters are adjusted based on labeled data from the downstream tasks. This process is

often referred to as a feature-based approach, as the pre-trained representations incorporate context from both directions (left-to-right and right-to-left). Figure 3 illustrates the input representation in the BERT model. The input embeddings are the sum of the token embeddings, the segmentation embeddings, and the position embeddings.

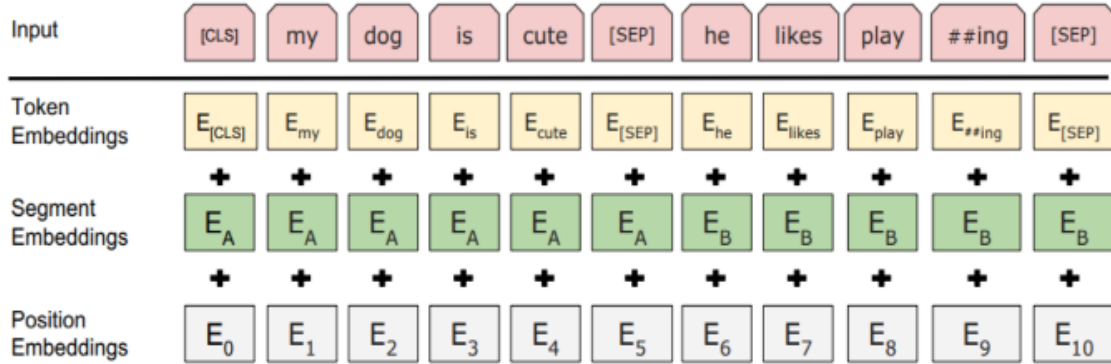


Figure 3: BERT Input Representation

The BERT model introduces a special classification token, [CLS], at the beginning of every sentence and uses the [SEP] token to differentiate between sentences. To further assist the model in distinguishing between sentences, a learned embedding is added to each token, indicating whether it belongs to sentence A or B. In this representation, the input embedding E is assigned to the first hidden vector of the [CLS] token as $C \in \mathbb{R}^H$. At the final hidden layer, the i -th input token is represented as $T_i \in \mathbb{R}^H$.

BERT has been successfully deployed in a wide range of natural language processing applications, yielding impressive performance, particularly in NLP-related

tasks [45]. Several variations of BERT have emerged, including BERTje [47], DistilBERT [48], BioBERT [49], Clinical BERT [50], AlphaBERT [51], AraBERT [52], RoBERTa [53], DeBERTa [54], and MobileBERT [55]. These variants have been successfully applied to various tasks such as classification, prediction, and NLP analysis. Figure 4 presents some notable BERT variants.

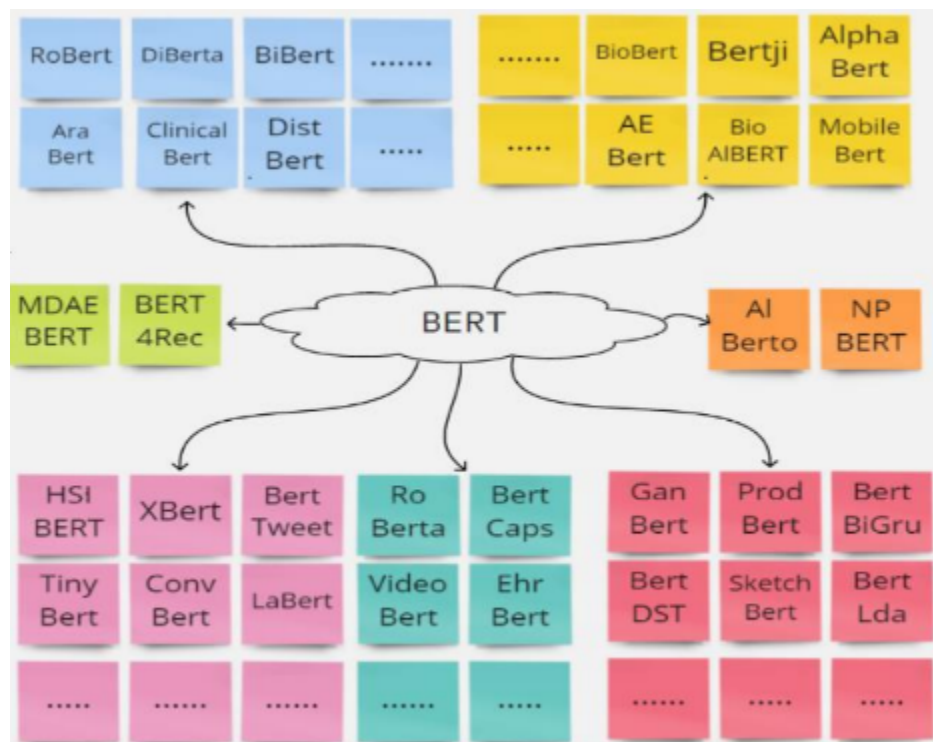


Figure 4: Various Types of BERT

2.8 Sentiment Analysis of Saudi Dialect

Advances in Saudi Dialect Sentiment Analysis

Significant advancements have been made in understanding the diversity of the Saudi dialect through the analysis of its unique linguistic characteristics using NLP methods. Researchers have achieved this by manually annotating conversations on social media platforms. Skilled annotators identify and categorize language-specific vocabulary, expressions, and grammatical structures. Although this method ensures high-quality annotations, it is also time-consuming and costly. It requires the collaboration of multiple teams of annotators. Al-Twairsh et al. introduced a collection of Arabic tweets labeled for sentiment analysis, focusing specifically on MSA and the SD. This corpus comprises 17,573 tweets, all manually annotated with four sentiment labels [56]. Similarly, Afnan Bayazed et al. developed a classified corpus of 4,180 multi-dialectal Saudi tweets for sentiment analysis in Arabic. Five native speakers annotated the tweets in two phases: first based on Saudi regions and then by sentiment [57]. Elshakankery, Fayek, and Farouk created a manually annotated dataset for sentiment analysis of Saudi tweets, known as LASTD, which includes more than 15,000 Arabic tweets labeled as positive, negative, and neutral [58]. A study by Alqarafi et al. developed an annotated sentiment corpus for the SD using data extracted from Twitter, which underwent thorough manual review and annotation [59]. Table 2 presents an overview of sentiment analysis evaluation tasks in the SD.

Table 2 - Summary of SD Datasets

Reference	Size	Evaluation Corpus	Evaluation Results
AraCust [3/42]	20,000 Tweets	Manually annotated sentiment analysis	Average F1-score = 89%
Saudi Corpus [3/43]	12,000 Tweets	Manually annotated for sentiment analysis	Accuracies ranging from 81% to 82.4%
SS2030 [3/44]	4,214 Tweets	Manually annotated for sentiment analysis	Accuracy of 89.83% using the SVM Model
SaudiShopInsights [3/45]	4,142 Reviews	Latent Dirichlet Allocation (LDA); A topic modeling was used	F1-score of 91.88%

Approaches to Analyzing the SD

The analysis of the Saudi dialect employs a variety of general approaches to comprehend its linguistic complexities and subtleties. These foundational methods pave the way for more advanced techniques, including machine learning (ML), deep learning (DL), and explainable AI (XAI) approaches. The analysis of the SD involves several techniques, such as DL-based sentiment analysis, psychophonetic methodologies for dialect identification, and perceptual dialectology methods [60], [61]. Research exploring language attitudes toward the SD has revealed underlying language ideologies and stereotypes associated with the three main regional dialects in Saudi Arabia. The study

utilized two primary methodologies. The first one was Verbal Guise Technique (VGT) which displayed voice samples. The second one was a semantic differential scale to assess adjectives on a continuum with opposing semantic labels [62]. These investigations enhance the understanding of the language environment and the application of sophisticated technologies in analyzing the SD.

Sentiment Classification Techniques

In analyzing sentiment in SD tweets, classification methods such as multinomial Naïve Bayes (MNB), support vector machines (SVM), and logistic regression classifiers have been utilized. These methods demonstrate the application of probabilistic models in deciphering the emotions and opinions conveyed in the SD [63]. Aseri et al. conducted a study in which they created classifiers using specific NLP models like logistic regression, MNB, and SVM to accurately differentiate between MSA and the SD at the sentence level. The models exhibited exceptional accuracy, with the MNB classifier outperforming the others [64]. Aftan and Shah investigated customer feedback prediction for Saudi telecom companies by analyzing Arabic Twitter datasets. They compared the performance of the AraBERT model with convolutional neural networks (CNN) and recurrent neural networks (RNN). The findings revealed that the AraBERT model surpassed both CNN and RNN models in accurately predicting customer satisfaction levels in the Saudi telecom industry, particularly with datasets from Mobily and STC [65]. Alqurashi's research applied character-level models to effectively identify various Arabic dialects, including those spoken in Saudi Arabia. This study employed classical

machine learning algorithms, such as logistic regression and support vector classification. It demonstrates greater accuracy compared to a deep learning character-based model [66]. These works highlight the efficacy of mathematical and probabilistic models in examining language patterns within the SD.

Ongoing Research and Phonetic Analysis

Ongoing research focuses on utilizing rule-based phonetics, morphological rule systems, and syntax parsing to analyze the SD. One approach for identifying dialects in Saudi Arabia employs socio-phonetic techniques to investigate the geographic distribution of various dialects in the region [60]. Furthermore, the application of specific NLP models for rule-based phonological models and morphological rule systems is a prevalent method in linguistic analysis. However, tailored applications for the SD may require additional research. Examining language attitudes in Saudi Arabia has provided valuable insights into the categorization of the SD, potentially aiding the advancement of linguistic analytic techniques [67]. Although data on the use of these specific methods for analyzing the SD remains limited, the general application of rule-based phonetics and morphological rule systems in linguistic analysis provides a foundation for their potential application in studying the SD.

2.9 Machine Learning Techniques in Saudi Dialect Analysis

Multiple machine learning methods have been employed to study the SD, including logistic regression, MNB, and SVM. For example, researchers utilized logistic

regression to analyze Saudi Arabian tweets for indicators of depression, achieving an accuracy rate of 82%. This study highlights the importance of leveraging social media data, particularly from Twitter, to identify depression within Arabic-speaking populations [68]. Techniques such as stochastic gradient descent, various Naïve Bayes models, and support vector classification have been applied to detect specific Arabic language dialects, including the SD. In addition, multilayer perceptrons (MLPs) have been utilized in several studies to evaluate the different dialects spoken in Saudi Arabia. One study employed MLP models to classify Arabic dialects by analyzing acoustic waves. By utilizing Mel-Frequency Cepstrum Coefficients (MFCC) features, the study found that K-nearest neighbor models were accurate 76% of the time, while MLP models achieved 34% accuracy [69]. Another investigation into sentiment analysis of Arabic texts discovered that MLP models performed exceptionally well in both Arabic and English sentiment analysis. Furthermore, Arabic SD speech data was used to train MLP models for emotion recognition, with SVM and MLP models achieving the highest accuracy at 77.14% [70]. These findings illustrate the effectiveness of deep learning techniques in evaluating the SD and enhancing the performance of traditional machine learning algorithms.

2.10 Hybrid Approaches to Sentiment Analysis

The analysis of the SD incorporates hybrid or enhanced techniques that merge specific NLP models with linguistic and machine learning methodologies to classify sentiment in dialectal Arabic. This approach integrates semantic modeling tailored to the

target domain with a multi-factor lexicon-based sentiment analysis. It also employs an innovative light-stemming technique to improve the stemming process for SD Arabic. A new hybrid technique for sentiment analysis has been specifically introduced for a dataset of the SD collected from Twitter. This methodology combines lexical and supervised learning techniques to analyze the sentiments expressed in Arabic tweets, particularly those in the SD [71].

In addition, a hybrid method is employed for sentiment analysis, incorporating domain knowledge in two ways: integrating problem domain semantic knowledge into the machine learning training feature set and using the results from lexicon-based sentiment classification to inform machine learning methodologies [72]. Another strategy combines a support vector machine (a machine learning predictive model) with Van Dijk's ideological square, a model of critical discourse analysis. This method is applied to cluster and understand the thoughts and attitudes expressed on Saudi Twitter. The study used computational techniques alongside critical discourse analysis to categorize and analyze the viewpoints of Saudi Twitter users concerning the actions of the Saudi General Entertainment Authority (GEA). It revealed two main factions with distinct ideological positions. The study aimed to demonstrate how computational linguistic techniques can be applied to analyze opinionated big data and identify the primary concepts and methods that create divisions in Saudi public opinion based on ideology [73].

CHAPTER III

METHODOLOGY

3.1 Introduction

This chapter will provide the proposed methodology for developing the SD sentiment analysis model. The architecture is explained in detail along with the use of two primary tools – the MARBERT and AraBERT models. The acquisition of X dataset, annotation layer, application of the GPT model for SD generation, the characteristics of the GPT generated dataset, and evaluation metrics are also explained in detail.

3.2 The Architecture of the Saudi Dialect Sentiment Analysis Model

To create a NLP model tailored for a low-resource language (LRL), the following approach was devised:

Step 1: Selecting an existing well-trained NLP model designed for a similar task but in a high-resource language (HRL) in order to leverage transfer learning. This model will be fine-tuned with data in the low-resource language to transform it into a model suited for that language.

Step 2: To minimize the effort and time needed for collecting and cleaning LRL data, generative AI is integrated into the process. The cleaned data gathered is fed into a generative AI model. That model then generates synthetic LRL data.

Step 3: Since the collected data may contain both LRL-specific and non-LRL-specific words and phrases, an annotator or filter is required to remove any non-LRL content. This step ensures the data used for fine-tuning the HRL model is purely in the LRL.

Step 4: The generative AI model needs to be trained with labeled data to ensure it accurately generates LRL content. The annotator/filter (created in Step 3) will label each generated data item to improve the model's accuracy.

Step 5: The same annotator/filter used for labeling in Step 4 will also be responsible for selecting LRL data to be included in the final dataset. This dataset will consist of both cleaned and generated LRL data. It will be used to fine-tune the HRL model into an LRL model.

Application of the Model to SD Sentiment Analysis

Applying this general framework to sentiment analysis in the Saudi dialect requires identifying a suitable high-resource model. Two potential models were identified: AraBERT, pre-trained on standard Arabic, and MARBERT, pre-trained on both standard Arabic and several Arabic dialects including the SD.

Although MARBERT seemed a promising candidate, its small size and subsequent fine-tuning for sentiment analysis produced unsatisfactory results. On the other hand, AraBERT is a larger language model. After fine-tuning, AraBERT proved to

be more effective for sentiment analysis in the SD. Therefore, AraBERT was selected as the HRL model for this task.

Despite MARBERT's inadequacy for sentiment analysis, it performed well as an annotator/filter for distinguishing LRL data after being fine-tuned with a small dataset of SD content. Hence, MARBERT was selected to fulfill the annotator/filter role. For generating synthetic Saudi dialect data, AraGPT2 was used. AraGPT2 is a GPT-2 model tailored for Arabic.

Data Collection and Preprocessing

Data was collected from X (formerly Twitter) to build the dataset. The data underwent preprocessing, which involved three primary steps: data cleaning, normalization, and tokenization. This process ensured the dataset was in a format suitable for use in our NLP models.

The approach to developing the Saudi dialect sentiment analysis model is based on fine-tuning a HRL model for LRL tasks. The mechanism is depicted in Figure 5. The next few sections further elaborate on the dataset. These include preprocessing techniques, examples, performance metrics, and the roles of MARBERT and AraBERT in the system.

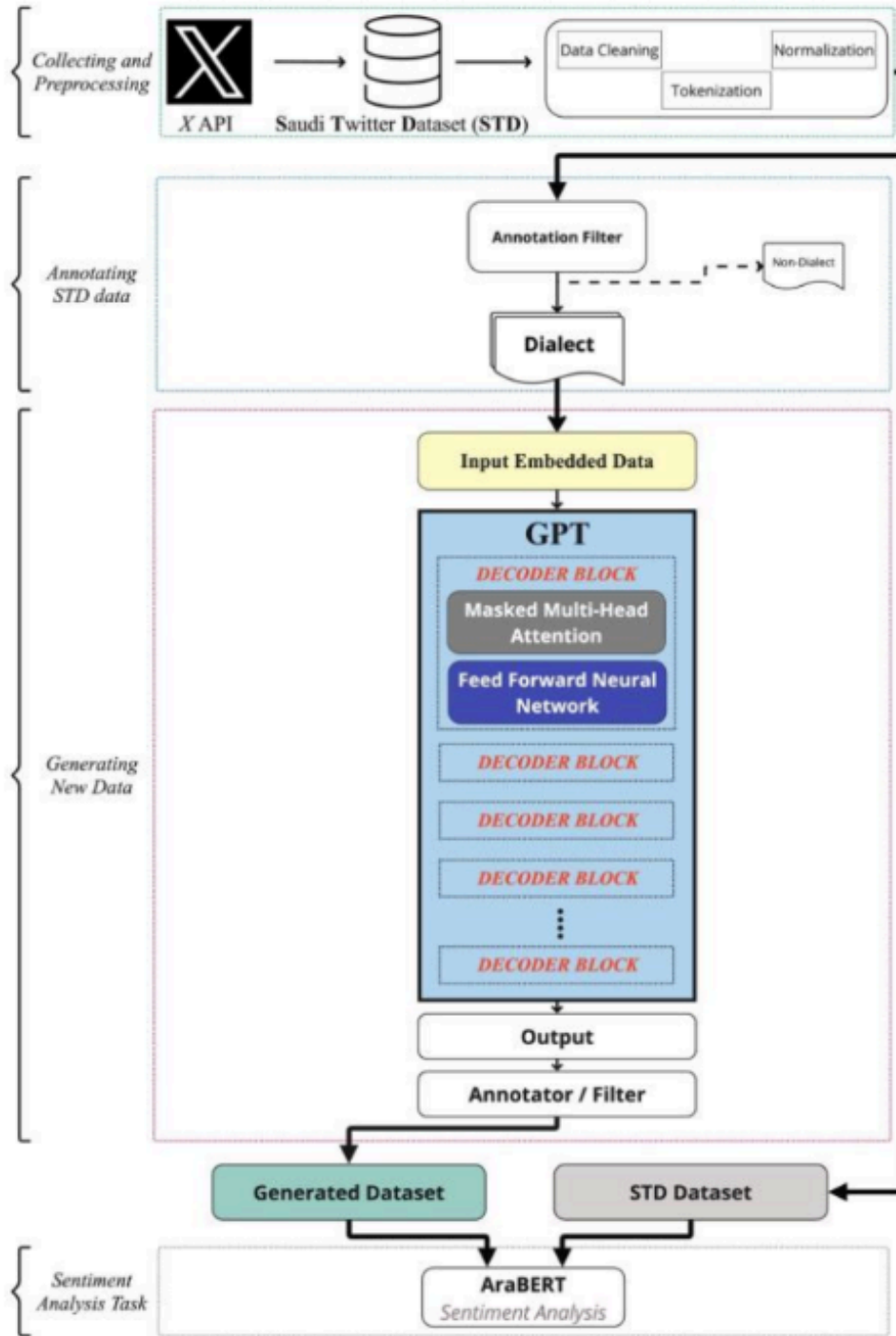


Figure 5: The Architecture of the SD Sentiment Analysis Model

3.3 X Dataset

Social Media Platforms and Data Collection in Saudi Arabia

In the Arab world, social media platforms such as X, Facebook, Instagram, and YouTube are widely used [44]. A study showed that most researchers focus on gathering Arabic text datasets from platforms like X and Facebook [74]. According to a report by the Saudi Communications, Space, and Technology Commission (CST), social media plays a crucial role in Saudi Arabia due to the high level of internet usage and the youthful demographic. As shown in Figure 6, X is the third most visited social media platform in the country, with over 25 million users [75].

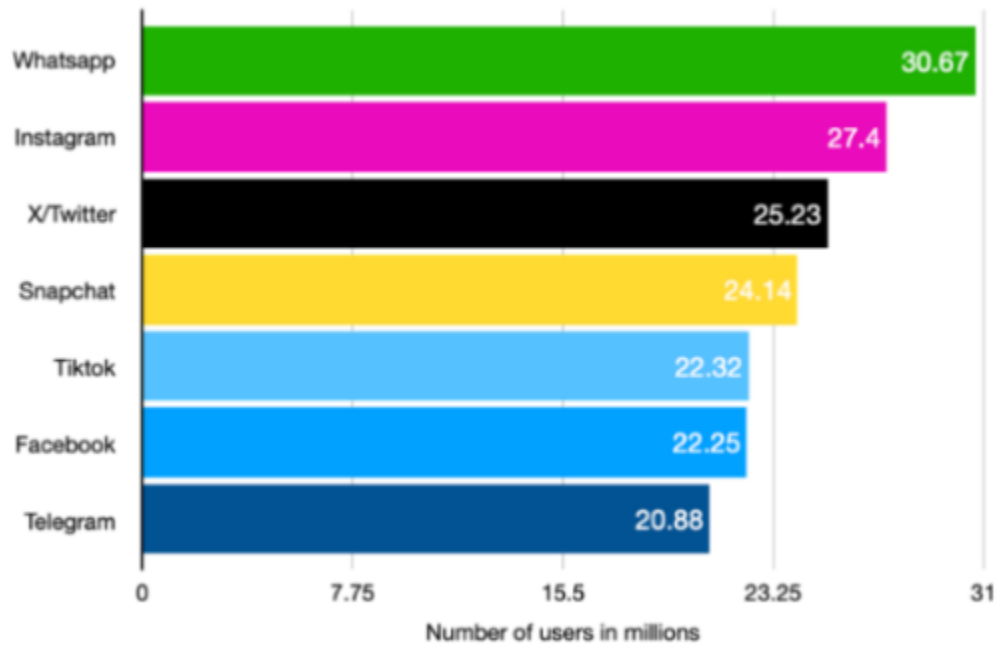


Figure 6: Most Visited Social Media Platforms in Saudi Arabia

Methodology for Collecting Tweets from Saudi Arabia

To collect tweets from Saudi Arabia using the X API v2, a structured approach was followed that involved several key steps. First, the working environment was set up by importing necessary libraries, creating authentication codes, generating URLs, and establishing a connection to the X API endpoint. The main goal was to extract tweets containing specific Arabic and dialect-related keywords, while ensuring that the tweets were geotagged from within Saudi Arabia.

Google Colab platform was used combined with NVIDIA's T4 GPU to enhance the performance of this task, particularly in high-performance computing applications. In addition, Tweepy, a Python library, was used to interact with X's API. The API access was authenticated using a Bearer token stored securely in environment variables. Keywords and phrases related to Saudi Arabia were identified, the country code for Saudi Arabia ('SA') was specified, and the language option was set to Arabic to filter the search results accordingly.

API URL Generation and Data Retrieval

A function to generate the API URL by incorporating relevant query parameters was designed. These parameters included expansions to collect more detailed tweet data and specific fields related to the search, such as the text and language of each tweet. A geo-bounding box was also added to limit results to Saudi Arabia. To comply with the X API's rate limits, a looping mechanism was implemented that sent requests in batches, with time intervals to prevent exceeding the request limit. By utilizing the 'next_token'

field in the API responses, the result pages were navigated, ensuring that the maximum number of relevant tweets were captured. Once valid responses were retrieved, the tweet data was stored in a CSV file for further analysis.

Data Collection and Processing of the Saudi Twitter Dataset

Between January and June 2024, approximately 50,000 Arabic tweets were collected. While many users did not provide useful geographical data, a notable portion did specify their city or region within Saudi Arabia. A thorough investigation was conducted to extract terms related to Saudi Arabia, including ‘KSA’, ‘Saudi’, the names of provinces and cities, and relevant hashtags. During the collection process, debugging print statements were used to verify the correctness of the generated URLs and to troubleshoot any errors in the API requests. This systematic approach allowed for collecting a substantial dataset of tweets that contained the selected keywords while ensuring compliance with X’s usage guidelines and resolving potential issues. Table 3 provides an overview of the collected dataset, named the Saudi Twitter Dataset (STD).

Table 3 - Description of the Saudi Twitter Dataset (STD)

Number of Tweets	50,000
Number of Words	1,320,788
Language	Arabic
Platform	X

3.4 Annotation of the X Dataset

I - Preprocessing the X Dataset for Semantic Analysis

Preprocessing is a crucial step in cleansing the dataset to reduce errors and enhance the accuracy of semantic analysis. This stage is particularly important for developing word embedding models, as the quality of preprocessing significantly affects the final results. It involves several steps, such as tokenization, removing URLs, punctuation, digits, unnecessary whitespace, user mentions, hashtags, and emojis. In this case, Python's NLTK (Natural Language Toolkit) library was utilized to handle the Arabic script by removing diacritics, eliminating Tatweel symbols, and correcting common spelling errors. This preprocessing ensured that the dataset was clean and suitable for training the MARBERT model. As a result, over 27,870 tweets were processed, containing a total of 379,974 words in Arabic. Table 4 provides an example of the preprocessing steps applied to the Saudi Twitter Dataset (STD) corpus.

Table 4 - Examples of Preprocessing Steps on STD Corpus

Preprocessing Steps	Example Before Processing	Example After Processing	Explanations
Removing URL	@LWHLH_ لا لا لا ياغزو ، دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا HTTPS://TWITTE R.COM/EKLMRE/ STATUS/9996848 53468626944	@LWHLH_ لا لا لا ياغزو ، دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	We excluded external website links from tweets to maintain focus on textual content.
Removing English letters, user mentions, and hashtags	@LWHLH_ لا لا لا ياغزو ، دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	لا لا لا ياغزو ، دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	In focusing on the Saudi Arabic dialect, we filtered English words from the tweets to ensure the model concentrated solely on Arabic content.
Removing punctuation	لا لا لا ياغزو ، دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	لا لا لا ياغزو دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	We removed punctuation marks from the tweets to standardize the text and aid in dialect classification.
Removing emojis	لا لا لا ياغزو دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	لا لا لا ياغزو دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	Emojis have been removed from the text to prevent them from affecting the language model's understanding.
Removing whitespace	لا لا لا ياغزو دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	لا لا لا ياغزو دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	Whitespaces has been removed to improve model performance.
Removing Arabic diacritics	لا لا لا ياغزو دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	لا لا لا ياهوه دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	We removed diacritics from the Arabic script to standardize the text and improve model performance.
Removing Arabic Tatweel	لا لا لا ياهوه دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	لا لا لا ياهوه دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا صحصحنا	We standardized the text by removing Tatweel characters, used in Arabic script for stylistic stretching, to avoid unnecessary variation.
Removing consecutive duplicate Arabic words	لا لا لا ياهوه دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا	لا ياهوه دايفه كنت بنام بس من كثر الضحك خلاص صحصحنا	We removed consecutive duplicate Arabic words to improve data quality and language pattern learning by the model.

II - Application of MARBERT Model for Saudi Dialect Annotation

A comprehensive list of terms and phrases in the SD and MSA was compiled to facilitate the task of Saudi dialect annotation. MARBERT model is a large masked language model (MLM) that focuses on both MSA and Arabic dialects. MARBERT was used for this task. MARBERT is built on the pre-trained BERT model which understands the contextual meaning of words by analyzing the words surrounding them. The model was applied using the Transformers library, which provides tools and pre-trained models for NLP. To process the text, a WordPiece tokenizer (designed for BERT models) was used to divide the text into tokens. This made it easier for the model to comprehend the data. The PyTorch package was used to train the text classification model.

Model Fine-Tuning and Evaluation

During the fine-tuning process, the MARBERT model learned to distinguish between dialectal and non-dialectal tweets by identifying patterns within supervised data. Hyperparameters, such as the learning rate, batch size, and number of training epochs, were optimized to enhance model performance. The dataset was split into 80% for training and 20% for testing to evaluate the model's generalization ability on unseen data. Metrics like accuracy, precision, recall, and F1-score were used to assess the model's performance. In addition, early stopping was implemented to prevent overfitting or underfitting. Error analysis helped identify recurring issues, such as code-switching between MSA and the SD. The evaluations on the unlabeled dataset allowed the model to predict whether each tweet was written in the Saudi dialect, and the results were saved as

an annotated dataset, ready for the next step of applying the GPT model.

3.5 Application of GPT Model for Saudi Dialect Generation

To enhance the dataset with dialectal text, a Generative Pre-trained Transformer (GPT) model was utilized on the annotated data [76]. The AraGPT2 model was selected for this task since it specializes in generating Arabic text [77]. The AraGPT2 model consists of 12 layers (blocks) of a transformer decoder architecture, which processes input data through these layers to produce meaningful output. With 1.46 billion parameters, AraGPT2 captures the complexity of Saudi dialect variations, allowing it to generate coherent and accurate text based on the annotated Saudi dialect tweets.

Data Processing with the GPT Model

The implementation was executed using Google Colab – a cloud platform that supports NVIDIA T4 GPU. After the necessary libraries were installed, the annotated dataset (which included only dialect-labeled tweets from the MARBERT model) was passed through the process. To handle large vocabulary and rare words, a Byte Pair Encoding (BPE) tokenizer was used to convert text into tokens. The token IDs were then transformed into dense vectors with 768 units in the embedding layer, which served as input to the model. A positional encoding layer was added to provide information about the position of each token in the sequence. To avoid overfitting during training, a random dropout layer of 0.1 was applied to deactivate some neurons. The tokens were processed through 12 transformer blocks, forming the core of the GPT model.

Attention Mechanism and Generation Process

A normalization layer was applied before the attention mechanism which stabilized and accelerated model training. The attention mechanism enabled the model to predict the next token based on the previous token in the sequence. Feed-forward networks were crucial in improving the model's learning capabilities by adding non-linearity. This helped the model capture the complexity of the SD. This process was iterated for each subsequent token until the end of the sequence. The output was then passed through a SoftMax layer, which converted the raw data into probabilities across the vocabulary. As shown in Figure 7, an additional annotation layer was incorporated into the pipeline to filter out irrelevant or poorly generated text to ensure that the generated text was relevant to the prompt [78].

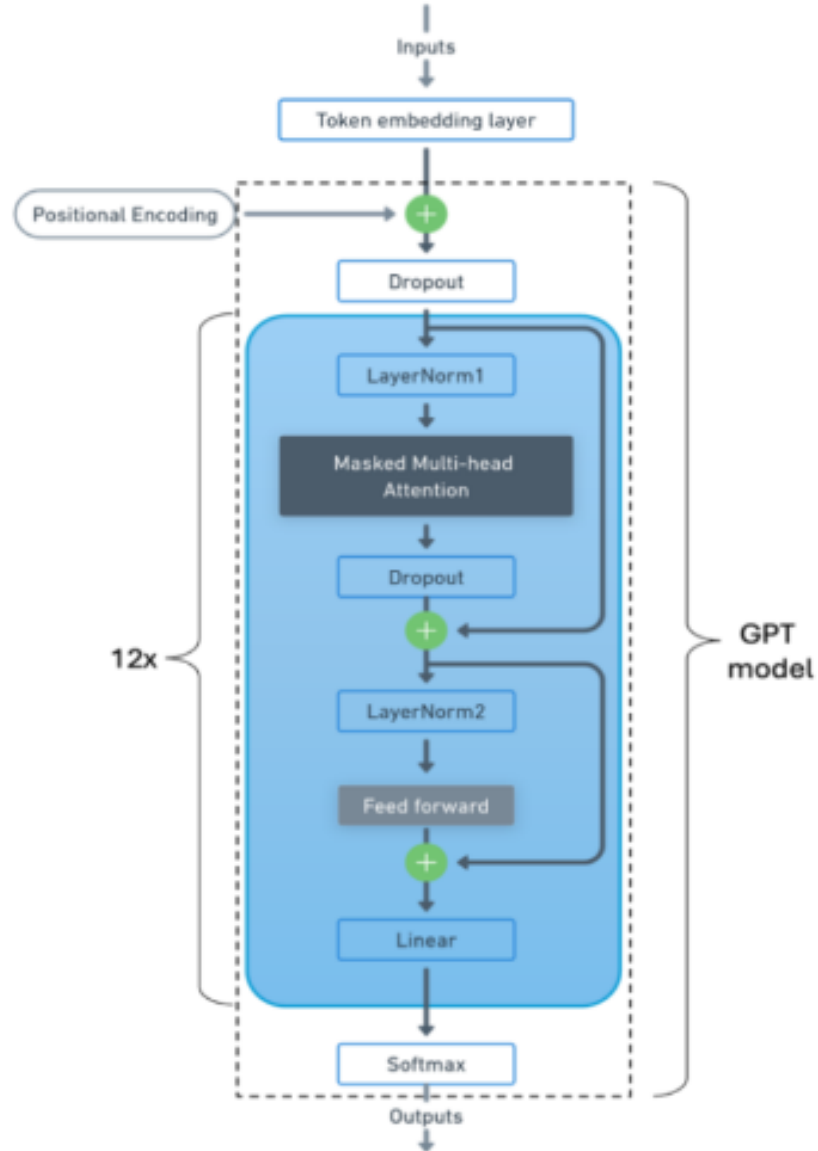


Figure 7: Architecture of AraGPT Model

Model Performance and Metrics

Model performance was evaluated using two metrics: perplexity and the BLEU (Bilingual Evaluation Understudy) score. Perplexity measures how well the model predicts the next token in a tweet. BLEU assesses the similarity between the generated

tweet and the original one [79]. By fine-tuning hyperparameters such as the number of epochs, batch size, and learning rate, and closely monitoring these metrics, the model's performance improved. This resulted in the generation of coherent and accurate tweets.

3.6 GPT Generated Dataset

Performance of the Fine-Tuned GPT Model

The fine-tuning of the AraGPT2 model on the Saudi Twitter Dataset (STD) yielded remarkable results. The model successfully generated human-like text while maintaining the coherent structure and syntactical nuances specific to the Saudi dialect. This outcome highlights the model's ability to learn and replicate linguistic patterns unique to the Saudi dialect from the provided training data. Table 5 provides a detailed description of the generated SD dataset.

Table 5 - Description of the Generated Dataset

Number of Generated Tweets	19,251
Total Characters in Generated Data	4,742,250
Average Characters per Generated Tweet	246.34
Total Words in Generated Data	950,627
Average Words per Generated Tweet	49.38

Evaluation of Generated Text Examples

Table 6 presents examples from the generated Saudi dialect dataset. In these examples, the original tweet prompts are highlighted in red, and the corresponding model-generated text is displayed in green. Following each example, the BLEU scores and perplexity scores are provided to evaluate the quality and accuracy of the generated text. The model demonstrated impressive capabilities – generating human-like responses, and also producing extended, coherent, and contextually appropriate content for the given tasks.

Table 6 - Examples of the Generated Saudi Dialect Dataset

Example 1	<i>Original Tweet</i>	هالصوره قصده لا تسوي نفسك قوطي ما تفهم
	<i>Generated Saudi dialect</i>	هالصوره قصده لا تسوي نفسك قوطي ما تفهم ولا تشخصن على الناس اللي انت قاعد تسمع كلام في تويتر، ترا حتى لو كان عندك حساب او شخص ياخذها من يوتيوب مثلا BLEU Score: 0.71 Perplexity Score: 17.34
Example 2	<i>Original Tweet</i>	شيء طبيعي لازم كان فيه بدائل استثمارية مثل ما قلت لك
	<i>Generated Saudi dialect</i>	شيء طبيعي لازم كان فيه بدائل استثمارية مثل ما قلت لك انت مثلا في شركات يعني لو عندك مشاريع زراعية او انتاج حيواني بس انا اتوقع هذا من التحذات اللي قاعده فيها هذه الشركات حتى الان على سبيل المثال BLEU Score: 0.85 Perplexity Score: 20.30
Example 3	<i>Original Tweet</i>	كلن له وجهة نظر
	<i>Generated Saudi dialect</i>	كلن له وجهة نظر ، حتى لو كان ما يفكر في نفسه عشان يعرف هو اللي بيصير له زي كذا، يعني ليه لما يشوفها بشكل مختلف عن غيره بيجوز BLEU Score: 0.88 Perplexity Score: 10.55
Example 4	<i>Original Tweet</i>	وش دخلك
	<i>Generated Saudi dialect</i>	وش دخلك في الموضوع؟ ترى هذا كلام من حقين الرجال وش تحسن فيه؟ يعني انت الحين قاعد تقول لنا لا تشخصن النساء BLEU Score: 0.92 Perplexity Score: 12.88
Example 5	<i>Original Tweet</i>	راعي السيارة غلطان ما فيها كلام
	<i>Generated Saudi dialect</i>	راعي السيارة غلطان ما فيها كلام الا صوته واضح انه راعي التباب مو ناق اشارة ولا غيره صاحب السيارة غلط في كلامه خصوصا لو كان الطريق يلتبيح ان هذا مخالفه من اليمين BLEU Score: 0.93 Perplexity Score: 08.20

3.7 Use of AraBERT Model for Sentiment Analysis

Introduction to the AraBERT Model

The AraBERT model is widely recognized in Arabic Natural Language Processing (NLP) for its effective application in sentiment analysis. AraBERT is a pre-trained language model specifically designed for the Arabic language, based on the BERT architecture. The model contains 12 encoder blocks, 768 hidden dimensions, 12 attention heads, and approximately 110 million parameters. Pre-trained on a vast corpus of Arabic texts – including social media, news stories, and Wikipedia entries – AraBERT is adept at capturing the complexities of the Arabic language [12]. That makes it highly suitable for various NLP tasks including sentiment analysis.

Comparative Analysis of AraBERT's Performance

A comparative analysis was conducted using the AraBERTv02 model across multiple datasets. These datasets included the Saudi Twitter Dataset (STD), the AraCust dataset, a generated dataset combining STD with AraCust, and another combination of the STD and AraCust datasets. The AraCust dataset focuses on analyzing customer feedback for Saudi telecom companies, containing 20,000 manually labeled tweets – 32% positive and 68% negative – collected from January to June 2017 [39].

Preprocessing and Data Preparation

Consistent preprocessing techniques were applied to all datasets to ensure uniformity. This process involved cleaning, normalizing, and tokenizing the datasets

before passing them to the AraBERT model. Following preprocessing, the datasets' size was reduced, and fine-tuning was conducted using uniform hyperparameters: a maximum sequence length of 128, batch size of 16, four training epochs, and the AdamW optimizer with a learning rate of 2e-5. Mixed precision data type "FP16" was used to optimize gradient computations. Class weights and early stopping were implemented to handle class imbalance and ensure optimal training performance.

Experimental Setup and Model Evaluation

The experiments were conducted on the Google Colab L4 GPU, with the datasets split into 80% for training and 20% for validation. The same validation set was used across all experiments to evaluate model performance. Several evaluation metrics were employed including precision, recall, and F1-scores for both positive and negative classes. Also, the overall accuracy metric and confusion matrix were used to visualize the model's classification performance across different classes.

Evaluation Metrics and Performance Calculation

The following metrics were used to assess the performance of the AraBERT model:

$$\textbf{Accuracy} = (\textbf{TP} + \textbf{TN}) / (\textbf{TP} + \textbf{FP} + \textbf{FN} + \textbf{TN}) \quad \text{Eq. 1}$$

$$\textbf{Precision} = \textbf{TP} / (\textbf{TP} + \textbf{FP}) \quad \text{Eq. 2}$$

$$\textbf{Recall} = \textbf{TP} / (\textbf{TP} + \textbf{FN}) \quad \text{Eq. 3}$$

$$\textbf{F1 - Score} = 2 * (\textbf{Precision} * \textbf{Recall}) / (\textbf{Precision} + \textbf{Recall}) \quad \text{Eq. 4}$$

where;

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

These metrics provided insight into the model's effectiveness in classifying sentiment in the Saudi dialect.

CHAPTER IV

RESULTS

4.1 Introduction

This chapter presents and discusses the simulation results obtained by MARBERT, AraGPT, XAI, and AraBERT of the datasets for annotation and sentiment analysis with various performance metrics. The objective is to evaluate the performance of each model in handling the complexities of the SD. The results highlight the strengths and limitations of these models. Visualizations from XAI were utilized to provide deeper insights into the model outputs.

4.2 MARBERT Model Performance

The MARBERT model was fine-tuned over five epochs using a batch size of 32 for both training and evaluation. When applied to labeled data, the model demonstrated its ability to differentiate between dialectal and non-dialectal phrases and words. The evaluation results were outstanding. The MARBERT model achieved an accuracy of 98%. Precision and recall metrics further confirmed its proficiency in identifying SD features. Table 7 presents the evaluation metrics. The minimal training loss is 0.0020, and the validation loss is 0.095. The precision, recall, and F1-score are 0.969, 0.989, and 0.979, respectively. This illustrates the model's capability to make accurate predictions.

Table 7 - MARBERT Model's Evaluation Metrics

Training Loss	0.0020
Validation Loss	0.095
Accuracy	0.981
Precision	0.969
Recall	0.989
F1-Score	0.979

The confusion matrix in Figure 8 highlights the model's effectiveness in distinguishing between dialectal and non-dialectal text. The MARBERT model learned the unique characteristics of the SD through the fine-tuning process. It enabled it to annotate tweets with precision.

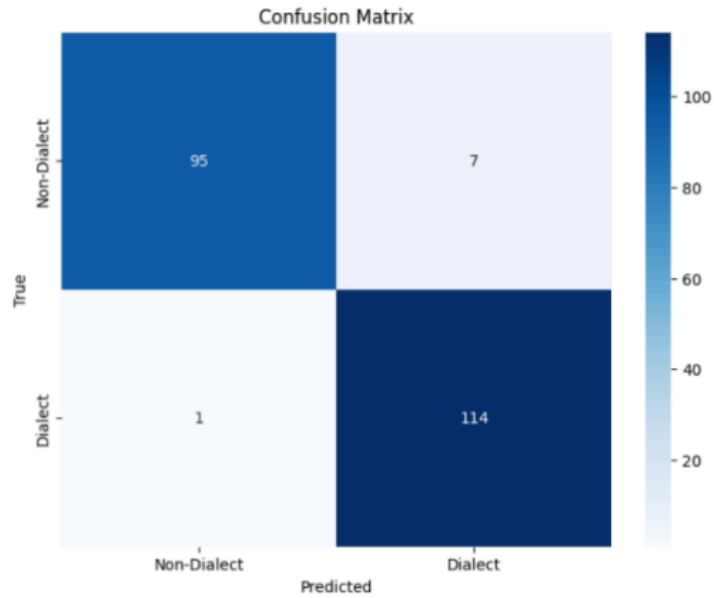


Figure 8: Confusion Matrix of the Annotated Dataset

After completing the annotation, 19,251 tweets (77.5% of the dataset) were classified as dialectal, while 8,620 tweets (22.5%) were classified as non-dialectal, as illustrated in Figure 9.

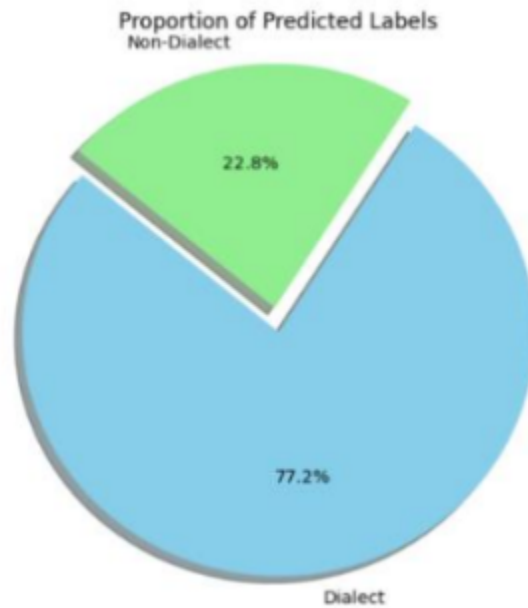


Figure 9: Predicted Label Distribution of the Annotated Dataset

4.3 eXplainable AI (XAI) on MARBERT

To better understand the complexity and outputs of the MARBERT model, eXplainable AI (XAI) tools were applied. XAI techniques aim to provide clarity and build trust in machine learning models by explaining their decision-making processes [80]. The objective of employing XAI was to improve the interpretability of MARBERT's predictions and performance. This was used for the annotated STD [81].

LIME was utilized to analyze and interpret the MARBERT model's predictions. By evaluating the keywords and phrases that influenced the model's decisions, LIME provided a detailed explanation of the model's performance, and offered insights into the factors that impacted its predictions.

LIME analyzed the keywords and phrases that significantly influenced the model's decision to classify tweets as either dialect or non-dialect. This evaluation highlights the specific words or phrases that impacted the model's predictions. Figure 10 provides examples of LIME explanations in tweets. It showcases the words or phrases that contributed to the classification along with scores representing each word's importance in the prediction.

In the first example, the original Arabic text is (يا واد قووم), When translated to English, it (*Hey man, get up*). But in the Saudi dialect, it means (*Stop lying*). In the second example the original Arabic text (tweet) is سؤال للجميع ماهي الشجرة إلى أصلها ثابت (وفرعها في السماء ؟). When translated to English and Saudi dialect, it means (*A question for everyone? What is the tree that has a firm root and its branches in the sky?*). The predicted class is not dialect.

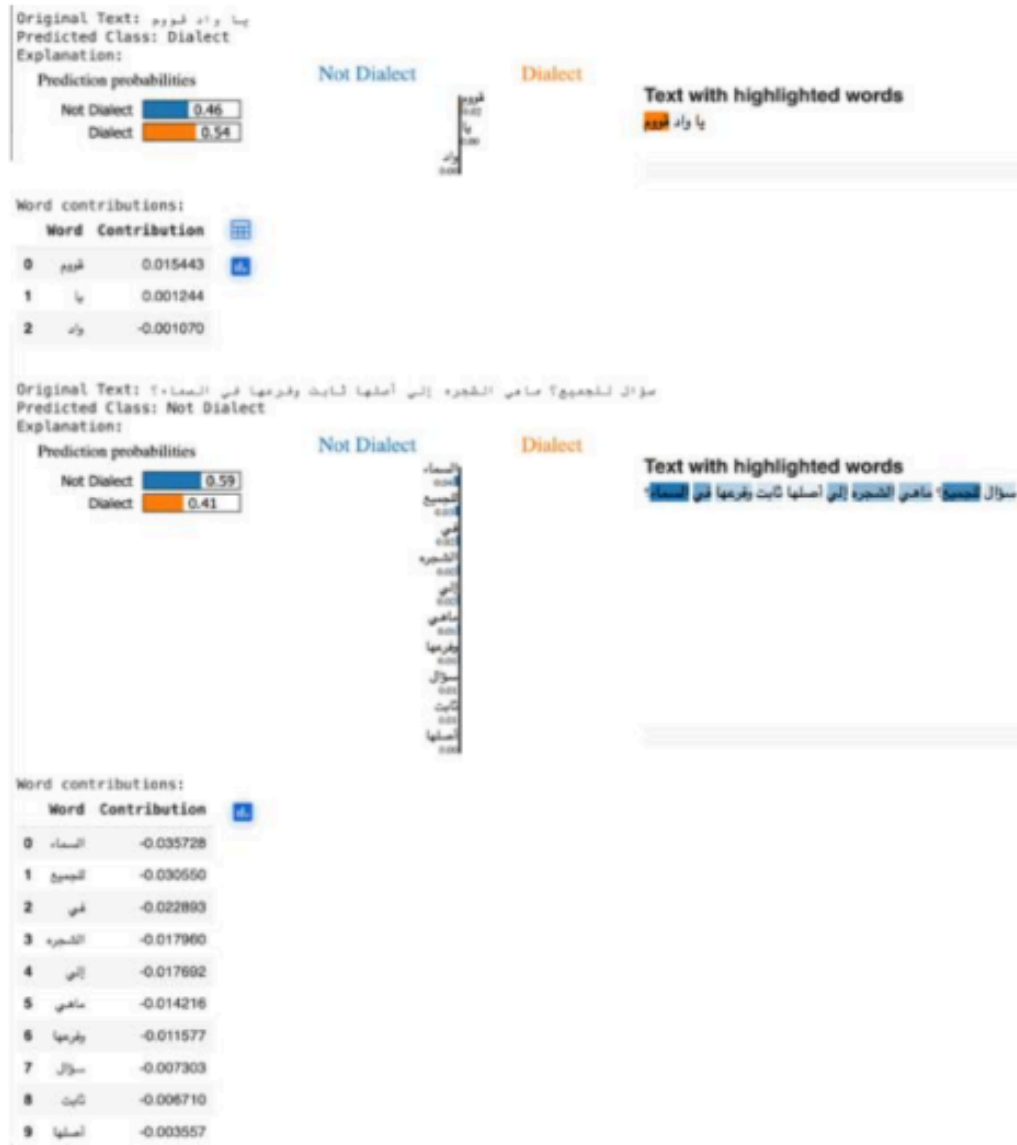


Figure 10: LIME Explanation of the MARBERT Model Prediction

4.4 AraGPT Model Performance

During the fine-tuning process of the AraGPT model, various hyperparameters and techniques were used to assess the quality of the generated data. The model was

trained for four epochs with a batch size of 16, a learning rate of $5e-5$, and a regularization technique set at 0.01 to prevent overfitting. The training process also involved initializing the trainer model and setting the generation command with a maximum sequence length of 128 tokens.

To balance creativity and coherence in the generated text, specific parameters were applied. The temperature was set to 0.7 which controls the randomness of the model's output. Higher values (e.g., 1.0) produce more diverse and unpredictable text, while lower values (e.g., 0.5) result in more deterministic output. The top_k value was set to 50 to limit the vocabulary and increase the model's focus. Nucleus sampling was applied with $top_p = 0.9$ to maintain diversity and avoid unlikely word choices. To prevent repetitive outputs, the `repetition_penalty` was set to 1.2 ensuring a more diverse range of generated words and phrases.

Perplexity Metric and BLEU Scores

The performance of the AraGPT2 model was evaluated using the perplexity metric and BLEU scores. Perplexity measures how well the model predicts the next word in a sequence. It showed an average value of 56.7. A lower perplexity indicates better model performance and higher confidence in its predictions for SD tweets. This result suggests that while the model has learned some patterns of the Saudi dialect, further improvements are possible since challenges remain in predicting dialect-specific tweets. The perplexity was calculated using the mathematical formula below:

$$PP(W) = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(w_i | w_1, w_2, \dots, w_{i-1}) \right) \quad \text{Eq. 5}$$

where;

W = the sequence of words in the test set. N : is the total number of words in the test set.

$P = (w_i | w_1, w_2, \dots, w_{i-1})$: is the probability of the word w_i given the previous words.

The BLEU score was used to measure the similarity between generated tweets and reference tweets. The BLEU score ranges from 0 to 1. Higher scores indicate better alignment between generated and reference texts. The average BLEU score of 0.65 suggests that the model generates coherent and somewhat accurate tweets in the Saudi dialect. However, further optimization is required to improve text generation quality and better mimic the dialect complexities.

4.5 AraBERT Performance on Sentiment Analysis

The AraBERT model was applied to evaluate the impact of various datasets on sentiment analysis tasks. A special emphasis was put on tweets written in the SD. In this experiment, the model's performance was primarily assessed using the AraCust dataset as a baseline to compare the effects of additional datasets on sentiment analysis.

According to the results presented in Table 8, the addition of the generated Saudi dialect dataset significantly improved AraBERT's performance when combined with

other datasets. However, the model struggled to effectively detect positive and negative sentiments when applied solely to the AraCust dataset. Interestingly, the combination of the generated dataset with AraCust allowed for a more diverse and straightforward dataset. This enhanced the model's learning and increased its accuracy from 89% to 97%.

Table 8 - AraBERT Model's Classification Report (after 1 Iteration)

Datasets	Evaluation Metrics	Precision	Recall	F1-Score
AraCust	Negative	0.98	0.88	0.93
	Positive	0.68	0.92	0.78
	Accuracy	0.89		
AraCust + Generated Dataset	Negative	0.98	0.98	0.98
	Positive	0.85	0.83	0.84
	Accuracy	0.97		
Saudi Twitter Dataset (STD)	Negative	0.79	0.65	0.71
	Positive	0.20	0.34	0.25
	Accuracy	0.58		
STD + AraCust	Negative	0.96	0.94	0.95
	Positive	0.90	0.93	0.92
	Accuracy	0.93		
STD + Generated Dataset	Negative	0.96	0.96	0.96
	Positive	0.94	0.92	0.93
	Accuracy	0.95		

AraBERT demonstrated balanced performance across both positive and negative sentiments. This highlights its suitability for sentiment analysis of SD tweets.

Nevertheless, when fine-tuned exclusively on the Saudi Twitter Dataset (STD), the model's performance was limited. This shows the importance of comprehensive training data. The merging of the STD and the generated dataset led to a higher performance improvement. The model's accuracy increased to 95%. This experiment showed the effectiveness of using synthetic data to complement real-world datasets, and in situations where class imbalance or data scarcity could hinder model performance. The generated dataset provides additional diversity and examples. It is critical to overcome these challenges and enhance AraBERT's overall performance across different datasets. The confusion matrix in Figure 11 illustrates the model's ability to accurately distinguish between positive and negative sentiments in SD tweets when trained on the combined datasets.

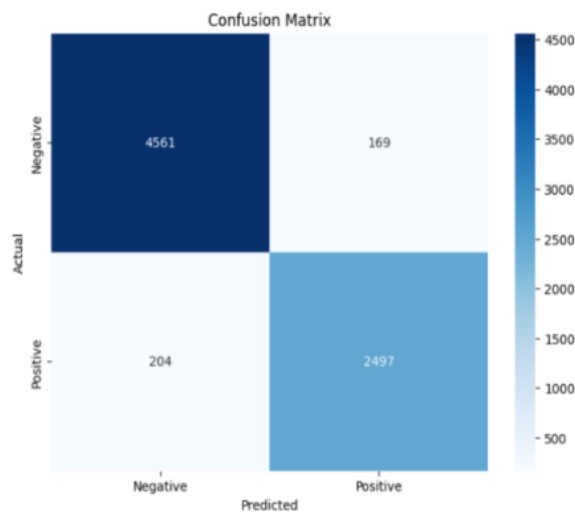


Figure 11: Confusion Matrix of AraBERT

Table 9 - Accuracy and F1-Scores (after 10 Iterations of Reshuffling)

Datasets	Accuracy (Average)	Accuracy (Standard Deviation)	F1-Score (Average)	F1-Score (Standard Deviation)
AraCust	0.8950	0.0830	0.8208	0.0321
AraCust + Generated Dataset	0.9647	0.0083	0.9215	0.0142
Saudi Twitter Dataset (STD)	0.8232	0.0250	0.6258	0.0426
STD + Generated Dataset	0.9201	0.0110	0.9001	0.0174
STD + AraCust	0.9274	0.0124	0.9037	0.0157
STD + AraCust + Generated Dataset	0.9474	0.0087	0.8927	0.0185

For AraCust, AraCust + Generated Dataset, STD + AraCust + STD + AraCust + Generated Dataset, these datasets were reshuffled ten times and split into training and testing sets using an 80-20 ratio. This method was used to ensure that the model evaluations are robust and reliable by averaging out any anomalies that may occur due to particular splits of the data. Reshuffling and splitting help simulate a more real-world scenario where the model would be expected to perform well across different subsets of data. The calculated averages and standard deviations for accuracy and F1 scores reflect the model's consistent performance across different shuffles and splits. Lower standard

deviations in these datasets suggest that the model's performance is stable regardless of how the data is divided.

For STD & STD + Generated Dataset, the metrics were calculated without the additional reshuffling or splitting into training and testing sets. Evaluations are based on a single partition of the data. The absence of reshuffling and splitting might result in higher variability in the model performance metrics as indicated by larger standard deviations compared to the reshuffled datasets.

Performance Metrics:

1. AraCust: Highest standard deviation in accuracy among the reshuffled datasets, indicating some fluctuations in model performance across different data splits.
2. AraCust + Generated Dataset: Exhibits the best overall performance with the highest accuracy and F1 scores and low standard deviations, demonstrating the beneficial impact of synthetic data on model consistency and effectiveness.
3. STD: Shows the lowest performance metrics and higher standard deviations, possibly due to the challenges inherent in the dataset such as dialect variations, and informal expressions.
4. STD + Generated Dataset: Improved accuracy and F1 scores compared to STD alone but still higher standard deviations suggesting that while synthetic data improves performance, the single-set evaluation might not provide a fully stable indicator of model reliability.

5. STD + AraCust + and STD + AraCust + Generated Dataset: Both show very good performance with relatively low standard deviations in accuracy, indicating that combining multiple real and synthetic datasets can lead to robust models capable of handling diverse data characteristics effectively.

The methodology of data preparation significantly impacts the stability and reliability of the performance metrics. Reshuffling and splitting the data typically provide more reliable indicators of how the model would perform in practical applications. This is evidenced by the lower standard deviations in those datasets compared to those evaluated on a single set basis. Table 9 highlights the effectiveness of different data combinations and the importance of rigorous evaluation methods in developing robust sentiment analysis models. Figure 12 shows the average scores of accuracy and F1-scores along with standard deviations for the different combination of datasets.

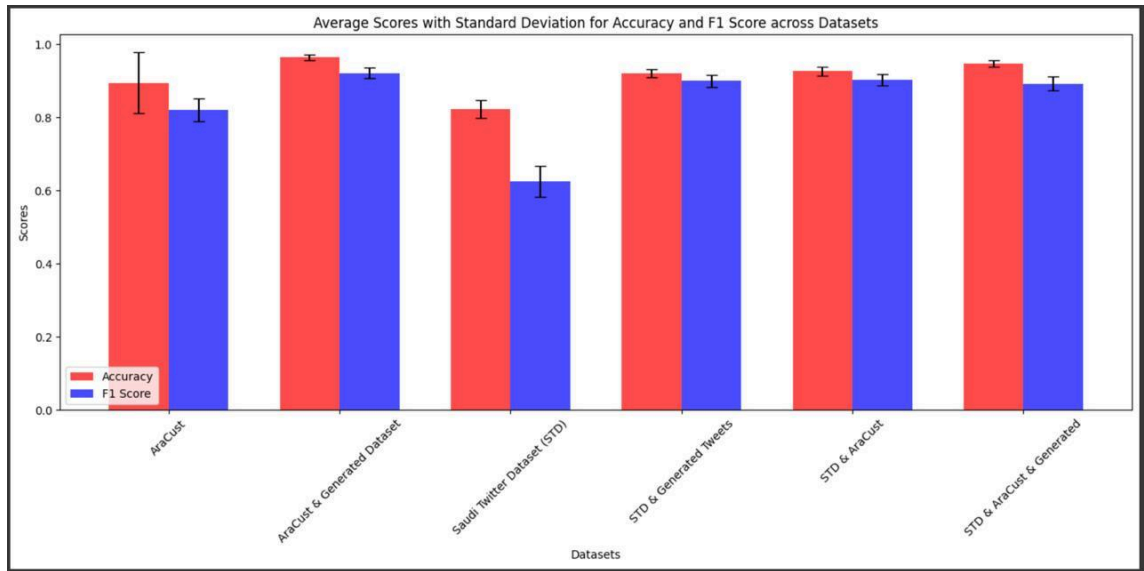


Figure 12: Chart of Average and Standard Deviation Scores for Datasets

CHAPTER V

CONCLUSION

5.1 Introduction

This chapter highlights the main findings and implications of the study on annotation and sentiment analysis of the SD using advanced deep learning models. Discussions on the limitations and sentiment analysis challenges of SD, opportunities in SD, and future research are also explored. Future opportunities lie in improving AI models through automation in corpus creation, enhancing cultural understanding, and enabling applications across various sectors such as healthcare, communication, and technology.

5.2 Main Findings and Implications

The annotation and sentiment analysis of Arabic present significant challenges. However, advancements in DL models have facilitated the processing of large datasets, making these tasks more manageable. The MARBERT model demonstrated effective application on a newly collected dataset for annotation. It achieved remarkable results with an F1 score and accuracy exceeding 98%. These findings provide reliable insights into the effectiveness of the results as visualized using the explainable AI (XAI) LIME model. It highlights the significance of each sentence structure for analysis from a dialectal perspective.

Moreover, the AraBERT model produced the highest performance results when the GPT-generated dataset was combined with the AraCust dataset. This combination led to a substantial increase in both accuracy and F1 score across all sentiment classes. The results obtained from this approach suggest that data augmentation through generated AI (such as the GPT model) can serve as a practical solution for enhancing NLP tasks. This is particularly relevant for LRLs and dialects like the SD. They can suffer from a scarcity of high-quality, large datasets.

Successful Survey of SD Research in NLP

A successful survey of SD research within the fields of NLP and Arabic Natural Language Processing (ANLP) over the past five years was performed. Despite the increasing interest in AI and language technologies, SD research has received relatively less attention compared to other dialects. One of the primary reasons for this gap is the unavailability of open datasets. AI researchers and developers in computer science encounter unique challenges when working with the Arabic dialect, particularly SD. These difficulties include limited access to extensive datasets, the nonconcatenative structure of the dialect, and the lack of short vowel orthographic representation in contemporary Arabic texts. Also, MSA requires a clear grammatical framework to indicate linguistic constituency which is complicated by the absence of case marking.

Need for Annotated Corpora

There is a critical need for annotated corpora to train and evaluate models specifically designed for SD. Manual annotation of SD corpora has been pursued to enhance reliability but this process is quite labor-intensive. It typically involves skilled individuals meticulously identifying and classifying language-specific lexicons, expressions, and syntactic patterns in social media discussions and other textual sources. Although manual annotation can produce high-quality datasets, it can also introduce bias and subjectivity, thus affecting the neutrality and objectivity of the corpus.

5.3 Limitations and Sentiment Analysis Challenges of SD

Challenges in Sentiment Analysis of Saudi Dialect

NLP researchers face numerous complex hurdles when working with Arabic dialects, including the SD. These challenges arise from both linguistic and data-related issues. A primary concern is the inherent diversity and complexity of Arabic dialects which exhibit significant regional and national variations. Due to differences in vocabulary, syntax, and morphology, models trained on one specific dialect may not perform effectively when applied to others [82]. For the SD, challenges include linguistic diversity, limited data availability and quality, and technical difficulties in processing dialectal variations [83]. The mix of Modern Standard Arabic (MSA) and colloquial forms within SD complicates language learning and translation efforts [84, 85]. The

limited availability of high-quality data on SD makes extensive research challenging, and NLP tasks require precise dialect identification., and accurate classifiers [86].

Linguistic Diversity and Standardization Issues

The SD has not been studied as extensively as other Arabic dialects, such as Egyptian or Levantine. Often, it is grouped with the broader Gulf dialect despite notable differences. One issue affecting NLP efforts is the lack of standardized spelling in SD. This poses challenges for developing semantic analyzers and part-of-speech (POS) taggers given that 80% of tweets are written in MSA [87]. Acquiring labeled data to train ML and DL models for sentiment analysis is difficult. Specialized dialects like SD require substantial labeled data. This is often hard to obtain [88].

Key Challenges in Saudi Dialect Analysis

1. Irregularity: Arabic dialects vary significantly across regions. Hence, it is difficult to standardize the language across sectors like business, media, education, and research. For example, pronunciation differences between individuals from Riyadh and Jeddah can cause confusion in professional settings. Research shows that interaction between the Najdi and Urban Hijazi varieties in Jeddah leads to changes in pronunciation, such as the affrication of /k/ and /g/ [89].

2. Diglossia: Diglossia involves the use of multiple dialects in everyday speech. It is a common occurrence across the Middle East. MSA is reserved for formal contexts like

media and education. This situation can make it challenging for speakers to switch between the spoken and written forms of the language.

3. Globalization: Increased communication between Arabic-speaking populations and the rest of the world has introduced new vocabulary and loanwords into Arabic dialects. This has been facilitated by technology. While this enriches the language, it also subjects dialects to international linguistic standards, potentially diminishing their uniqueness. In Saudi Arabia, globalization has modernized the economy in urban areas like Jeddah where English is gaining importance [90].

4. Rapid Urban Development: Urbanization has led to the concentration of people in cities, resulting in increased interaction among speakers of diverse linguistic backgrounds. This can lead to dialect homogenization, and potentially reduce linguistic diversity.

5. Mode of Learning: In many Arab countries, teaching MSA is prioritized over regional dialects. While this ensures proficiency in the official language, it can marginalize dialect speakers and hinder their linguistic development. Consequently, the number of native dialect speakers is declining due to educational practices that favor MSA.

6. Technological Difficulties: Although digital platforms can facilitate communication and language learning, the dominance of English on these platforms may limit the online presence of Arabic dialects. This makes it harder to preserve and develop these dialects in the digital era.

7. Limited Resources for Dataset and Tools: Unlike widely spoken languages like English, dialects such as SD have fewer resources, datasets, tools, and language models available for NLP study. This scarcity hampers the development of applications like machine translation, sentiment analysis, and named entity recognition. For example, building resources for low-resource languages (such as a dictionary for the Albanian language) poses similar challenges [91].

8. Linguistic Variation and Ambiguity: Arabic dialects including SD show variation in verb conjugation, noun morphology, and word order, leading to difficulties in NLP tasks that involve parsing and vocabulary analysis. The SD is also known for semantic ambiguity and polysemy, where words and phrases can have multiple meanings depending on the context. For example, the term “كتب” can mean “he wrote” or “books” depending on the vowel sounds. Resolving this ambiguity is crucial for accurate semantic role labeling, word sense disambiguation, and semantic analysis in NLP applications.

9. Limited Research on Saudi Dialect in NLP: Despite the potential for advanced AI methods to analyze Arabic dialects including SD, there is a significant gap in existing research. Few studies have focused on NLP applications for Arabic dialects. This suggests a need to motivate researchers to explore this area further. Increased interest in studying SD can lead to the development of more robust NLP techniques, facilitating better understanding and processing of this dialect.

5.4 Opportunities in SD Analysis

Enhancing Cultural Understanding and Communication

Studying the Saudi Dialect (SD) can significantly enhance cultural comprehension by providing a deeper understanding of Saudi Arabia's traditions and values. This will improve communication between Saudi citizens and individuals from other countries. It can lead to stronger international connections and partnerships. Proficiency in the SD also enhances work opportunities for individuals who want to effectively navigate the local business environment. Research and instructional efforts focused on analyzing the SD can lead to the development of language learning programs and materials tailored specifically for this dialect. The beneficiaries of such programs will benefit both Saudi and non-Saudi learners. For example, non-native individuals working in Saudi Arabia who have a thorough understanding of Saudi norms and traditions can build trust with their Saudi colleagues more easily. Successful collaborations and commercial relationships can be facilitated.

Opportunities in AI and NLP

The study of SD presents significant opportunities for activities related to dialect identification in NLP and downstream tasks in AI [86]. As the demand for AI and NLP technology continues to grow, analyzing the SD offers valuable insights into the cultural and historical context of the region. The enhanced understanding of the language and its speakers have numerous practical implications. In addition, a deep understanding of SD intricacies can lead to the development of more accurate and effective sentiment analysis

techniques. These approaches can be used to assess customers' emotions and opinions with greater precision and efficiency [91]

Advancing Technology Through Saudi Dialect Analysis

Developing Arabic corpora focused specifically on the SD is essential for advancing technological applications and social media analysis [92]. Effective models should leverage contextual information to disambiguate word meanings and understand intended messages. The use of unsupervised and semi-supervised learning techniques can also use unlabeled data, reducing the need for manual annotation. More favorable outcomes can be achieved by refining pre-existing models trained on similar languages or dialects and enhancing them with SD data. These strategies can improve the precision and effectiveness of NLP tasks specific to the SD.

Industry Demand for Saudi Dialect Proficiency

The integration of SD into NLP applications offers a wide range of opportunities across various industries. Companies in the technology and IT sectors are actively seeking professionals knowledgeable in SD to develop software solutions for the local market. Expertise in SD enables the connection between technological solutions and cultural understanding. Successful implementation of tools can be facilitated such as dialect translation, voice translation, chatbots, and applications designed for learning and practicing the SD. There is a high demand for individuals proficient in SD communication in the telecommunications and hospitality industries. Understanding the language and its cultural nuances is critical for ensuring a positive experience for visitors.

Potential Applications in Healthcare and Communication

The use of SD in NLP applications can transform how information is exchanged and interpreted across various contexts. For example, analyzing the emotions expressed by elderly patients in SD can help healthcare professionals adjust their communication style. There can be improved rapport, trust, and compliance with treatment plans. As a result, more effective and accurate communication between patients and healthcare providers can be attained. As illustrated in Figure 13, such applications underscore the potential of SD in revolutionizing the way information is processed and understood across sectors. Increased effectiveness of communication and services using SD will be a major asset for its users.



Figure 13: SD Opportunities by using NLP

5.5 Future Research

There is a need for automation and efficiency in corpus creation. Advancements in NLP technology can lead to the development of more efficient and cost-effective methods for generating dialect-specific corpora. There is a growing need for algorithms capable of automatically detecting and labeling dialect-specific features across large volumes of textual data. These automated approaches have the potential to significantly reduce the reliance on manual annotation. As a result, the efficiency of corpus creation for the SD needs to be increased. Developing such technology would mark a significant step forward in addressing the challenges associated with SD research. It will expand the capabilities of AI systems to process and understand this particular dialect.

REFERENCES

- [1] Audrey Azoulay, “United Nations Educational, Scientific and Cultural Organization (UNESCO)”
- [2] K. of S. Arabia. S. C. General Authority for Statistics, “General Authority for Statistics, Kingdom of Saudi Arabia. Saudi Census 2020”, 2020. Accessed: Sep. 03, 2024. [Online]. Available: <https://database.stats.gov.sa/home/report/3868>
- [3] O. F. Zaidan and C. Callison-Burch, “Arabic dialect identification”, *Computational Linguistics*, vol. 40, no. 1, pp. 171–202, 2014.
- [4] N. Habash, O. Rambow, M. Diab, and R. Kanjawi-Faraj, “Guidelines for annotation of Arabic dialectness” in *Workshop on Arabic and its local languages*, 2008.
- [5] T. Bent, E. Atagi, A. Akbik, and E. Bonifield, “Classification of regional dialects, international dialects, and non-native accents”, *J Phon*, vol. 58, pp. 104–117, 2016.
- [6] L. H. Baniata and S. Kang, “Transformer Text Classification Model for Arabic Dialects That Utilizes Inductive Transfer” *Mathematics*, vol. 11, no. 24, p. 4960, 2023.
- [7] A. A. Al Shamsi and S. Abdallah, “A systematic review for sentiment analysis of arabic dialect texts researches” in *Proceedings of International Conference on Emerging Technologies and Intelligent Systems: ICETIS 2021 Volume 2*, Springer, 2022, pp. 291–309.
- [8] M. El-Masri, N. Altrabsheh, H. Mansour, and A. Ramsay, “A web-based tool for Arabic sentiment analysis”, *Procedia Comput Sci*, vol. 117, pp. 38–45, 2017.
- [9] L. Lulu and A. Elnagar, “Automatic Arabic dialect classification using deep learning models”, *Procedia Comput Sci*, vol. 142, pp. 262–269, 2018.
- [10] N. Elhassan et al., “Arabic Sentiment Analysis Based on Word Embeddings and Deep Learning”, *Computers*, vol. 12, no. 6, 2023, doi: 10.3390/computers12060126.

- [11] M. Abdul-Mageed, A. R. Elmadany, and E. M. B. Nagoudi, “ARBERT & MARBERT: Deep bidirectional transformers for Arabic” in *ACL-IJCNLP 2021 - 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Proceedings of the Conference*, 2021. doi: 10.18653/v1/2021.acl-long.551.
- [12] W. Antoun, F. Baly, and H. Hajj, “Arabert: Transformer-based model for arabic language understanding” *arXiv preprint arXiv:2003.00104*, 2020.
- [13] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding”, *arXiv preprint arXiv:1810.04805*, 2018.
- [14] Pluto Labs Inc., “Finding R&D Trends and Experts Made Simple,” [Online]. Available: <https://www.scinapse.io>.
- [15] L. M. Alhazmi and A. A. Alfaifi, “Dialect identification in Saudi dialects: A socio-phonetic approach”, *Journal of Language and Linguistic Studies*, vol. 18, no. 1, pp. 820–835, 2022.
- [16] A. Al-Thubaity, Q. Alqahtani, and A. Aljandal, “Sentiment lexicon for sentiment analysis of Saudi dialect tweets”, *Procedia Comput Sci*, vol. 142, pp. 301–307, 2018.
- [17] A. A. Noman, U. H. Akter, T. H. Pranto, and A. K. M. B. Haque, “Machine Learning and Artificial Intelligence in Circular Economy: A Bibliometric Analysis and Systematic Literature Review,” 2022.
- [18] E. Haugen, “Dialect, language, nation 1,” *American Anthropologist*, vol. 68, pp. 922–935, 1966.
- [19] Statista Inc., “Natural Language Processing - Global | Market Forecast,” [Online]. Available: <https://www.statista.com/outlook/tmo/artificial-intelligence/natural-language-processing/worldwide>.
- [20] R. Coleman, “Dialectal variation in republican latin, with special reference to praenestine,” *Proceedings of the Cambridge Philological Society*, vol. 36, 1990. doi: 10.1017/S0068673500005204.
- [21] C. Holes, *Modern Arabic: Structures, Functions, and Varieties*. Georgetown University Press, 2004.

- [22] M. Ennaji, "Aspects of multilingualism in the maghreb," *International Journal of the Sociology of Language*, vol. 87, 1991. doi: 10.1515/ijsl.1991.87.7
- [23] K. Brustad and E. Zuniga, "Levantine Arabic," in *The Semitic Languages*, 2019.
- [24] C. Holes, *Gulf Arabic*. Routledge, 2003.
- [25] K. Versteegh, "Linguistic contacts between Arabic and other languages," *Arabica*, vol. 48, 2001. doi: 10.1163/157005801323163825.
- [26] A. Assiri, A. Emam, and H. Al-Dossari, "Towards enhancement of a lexicon-based approach for Saudi dialect sentiment analysis," *Journal of Information Science*, vol. 44, 2018. doi: 10.1177/0165551516688143
- [27] T. Prochazka, *Saudi Arabian Dialects*. Routledge, 2013.
- [28] A. M. A. Alqadasi, M. S. Sunar, S. Turaev, R. Abdulghafor, M. S. Hj Salam, A. A. S. Alashbi, A. A. Salem, and M. A. H. Ali, "Rule-Based Embedded HMMs Phoneme Classification to Improve Qur'anic Recitation Recognition," *Electronics (Switzerland)*, vol. 12, 2023. doi: 10.3390/electronics12010176.
- [29] C. Holes, *Colloquial Arabic of the Gulf and Saudi Arabia*. Routledge & Paul, 1984.
- [30] K. Walters, "Diglossia, linguistic variation, and language change in Arabic," *Amsterdam Studies in the Theory and History of Linguistic Science Series 4*, pp. 157–200, 1996.
- [31] A. Alsudais, W. Alotaibi, and F. Alomary, "Similarities between Arabic dialects: Investigating geographical proximity," *Inf. Process. Manag.*, vol. 59, p. 102770, 2022.
- [32] A. Al-Essa, "When Najd meets Hijaz: Dialect contact in Jeddah," 2009.
- [33] R. Tachicart, K. Bouzoubaa, and H. Jaafar, "Lexical differences and similarities between Moroccan dialect and Arabic," in *Colloquium in Information Science and Technology*, CIST, 2016.
- [34] O. F. Zaidan and C. Callison-Burch, "Arabic dialect identification," *Comput. Linguist.*, vol. 40, pp. 171–202, 2014.

- [35] M. W. Cowell, *A Reference Grammar of Syrian Arabic (Based on the Dialect of Damascus)*. Arabic Series, Number Seven, 1964.
- [36] N. Habash, F. Eryani, S. Khalifa, O. Rambow, D. Abdulrahim, A. Erdmann, R. Faraj, W. Zaghouani, H. Bouamor, N. Zalmout, S. Hassan, F. Al-Shargi, S. Alkhereyf, B. Abdulkareem, R. Eskander, M. Salameh, and H. Saddiki, “Unified guidelines and resources for Arabic dialect orthography,” in *LREC 2018 - 11th International Conference on Language Resources and Evaluation*, 2019.
- [37] F. Alshargi, S. Dibas, S. Alkhereyf, R. Faraj, B. Abdulkareem, S. Yagi, O. Kacha, N. Habash, and O. Rambow, “Morphologically annotated corpora for seven Arabic dialects: Taizi, Sanaani, Najdi, Jordanian, Syrian, Iraqi, and Moroccan,” in *ACL 2019 - 4th Arabic Natural Language Processing Workshop, WANLP 2019 - Proceedings of the Workshop*, 2019.
- [38] R. M. Alahmary, H. Z. Al-Dossari, and A. Z. Emam, “Sentiment analysis of saudi dialect using deep learning techniques” in *ICEIC 2019 - International Conference on Electronics, Information, and Communication*, 2019.
doi: 10.23919/ELINFOCOM.2019.8706408.
- [39] L. Almuqren and A. Cristea, “AraCust: a Saudi Telecom Tweets corpus for sentiment Analysis”, *PeerJ Comput Sci*, vol. 7, p. e510, May 2021, doi: 10.7717/peerj-cs.510.
- [40] S. Aftan and H. Shah, “Using the AraBERT Model for Customer Satisfaction Classification of Telecom Sectors in Saudi Arabia”, *Brain Sci*, vol. 13, no. 1, 2023, doi: 10.3390/brainsci13010147.
- [41] A. Adadi and M. Berrada, “Peeking Inside the Black-Box: A Survey on Explainable Artificial Intelligence (XAI)”, *IEEE Access*, vol. 6, 2018, doi: 10.1109/ACCESS.2018.2870052.
- [42] M. Atabuzzaman, M. Shajalal, M. B. Baby, and A. Boden, “Arabic Sentiment Analysis with Noisy Deep Explainable Model”, in *ACM International Conference Proceeding Series*, 2023. Doi: 10.1145/3639233.3639241.

- [43] Y. Abdelwahab, M. Kholief, and A. A. H. Sedky, “Justifying Arabic Text Sentiment Analysis Using Explainable AI (XAI): LASIK Surgeries Case Study”, *Information (Switzerland)*, vol. 13, no. 11, 2022, doi: 10.3390/info13110536.
- [44] A. A. Al Shamsi and S. Abdallah, “A Systematic Review for Sentiment Analysis of Arabic Dialect Texts Researches” in *Lecture Notes in Networks and Systems*, 2022. doi: 10.1007/978-3-030-85990-9_25.
- [45] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2018.
- [46] U. Özdil, B. Arslan, D. E. Taşar, G. Polat, and Ş. Ozan, “Ad Text Classification with Bidirectional Encoder Representations,” in *2021 6th International Conference on Computer Science and Engineering (UBMK)*, 2021, pp. 169–173. doi:10.1109/UBMK52708.2021.9558966.
- [47] W. de Vries, A. van Cranenburgh, A. Bisazza, T. Caselli, G. van Noord, and M. Nissim, “Bertje: A dutch bert model,” *arXiv preprint arXiv:1912.09582*, 2019.
- [48] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
- [49] J. Lee et al., “BioBERT: a pre-trained biomedical language representation model for biomedical text mining,” *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.
- [50] E. Alsentzer et al., “Publicly available clinical BERT embeddings,” *arXiv preprint arXiv:1904.03323*, 2019.
- [51] Y.-P. Chen, Y.-Y. Chen, J.-J. Lin, C.-H. Huang, and F. Lai, “Modified bidirectional encoder representations from transformers extractive summarization model for hospital information systems based on character-level tokens (AlphaBERT): development and performance evaluation,” *JMIR medical informatics*, vol. 8, no. 4, p. e17787, 2020.
- [52] W. Antoun, F. Baly, and H. Hajj, “Arabert: Transformer-based model for arabic language understanding,” *arXiv preprint arXiv:2003.00104*, 2020.
- [53] Y. Liu et al., “Roberta: A robustly optimized bert pre-training approach,” *arXiv preprint arXiv:1907.11692*, 2019.

- [54] P. He, X. Liu, J. Gao, and W. Chen, “Deberta: Decoding-enhanced bert with disentangled attention,” *arXiv preprint arXiv:2006.03654*, 2020.
- [55] Z. Sun, H. Yu, X. Song, R. Liu, Y. Yang, and D. Zhou, “Mobilebert: a compact task-agnostic bert for resource-limited devices,” *arXiv preprint arXiv:2004.02984*, 2020.
- [56] N. Al-Twairish, H. Al-Khalifa, A. Al-Salman, and Y. Al-Ohali, “AraSenTi-Tweet: A Corpus for Arabic Sentiment Analysis of Saudi Tweets,” in *Procedia Computer Science*, 2017.
- [57] A. Bayazed, O. Torabah, R. AlSulami, D. Alahmadi, A. Babour, and K. Saeedi, “SDCT: Multi-Dialects Corpus Classification for Saudi Tweets,” *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, 2020. doi: 10.14569/IJACSA.2020.0111128.
- [58] K. Elshakankery, M. Fayek, and M. Farouk, “LASTD: A Manually Annotated and Tested Large Arabic Sentiment Tweets Dataset,” in *ACM International Conference Proceeding Series*, 2021.
- [59] A. Alqarafi, A. Adeel, A. Hawalah, K. Swingler, and A. Hussain, “A Semi-supervised Corpus Annotation for Saudi Sentiment Analysis Using Twitter,” in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 2018.
- [60] L. M. Alhazmi and A. A. Alfaifi, “Dialect identification in Saudi dialects: A socio-phonetic approach,” *J. Lang. Linguist. Stud.*, vol. 18, pp. 820–835, 2022.
- [61] R. M. Alahmary, H. Z. Al-Dossari, and A. Z. Emam, “Sentiment analysis of Saudi dialect using deep learning techniques,” in *ICEIC 2019 - Int. Conf. Electronics, Information, and Communication*, 2019.
- [62] L. M. Alhazmi, “Language attitudes in fast-growing societies: New insights in the dynamism dimension,” *Humanit. Soc. Sci. Commun.*, vol. 10, 2023. doi: 10.1057/s41599-023-01988-1.
- [63] A. A. Alfutamani and H. H. Al-Baity, “Emotional analysis of Arabic Saudi dialect tweets using a supervised learning approach,” *Intell. Autom. Soft Comput.*, vol. 29, 2021. doi: 10.32604/iasc.2021.016555.

- [64] Y. Aseri, K. Alreemy, S. Alelyani, and M. Mohanna, “Meeting Challenges of Modern Standard Arabic and Saudi Dialect Identification,” presented at the 2022.
- [65] S. Aftan and H. Shah, “Using the AraBERT Model for Customer Satisfaction Classification of Telecom Sectors in Saudi Arabia,” *Brain Sci.*, vol. 13, 2023. doi: 10.3390/brainsci13010147.
- [66] T. Alqurashi, “Applying a Character-Level Model to a Short Arabic Dialect Sentence: A Saudi Dialect as a Case Study,” *Appl. Sci. (Switzerland)*, vol. 12, 2022. doi: 10.3390/app122312435.
- [67] O. M. Aldosaree, “Language attitudes toward Saudi dialects,” California State University, Long Beach, 2016.
- [68] N. Al-Musallam and M. Al-Abdullatif, “Depression Detection Through Identifying Depressive Arabic Tweets From Saudi Arabia: Machine Learning Approach,” in *Proc. 2022 5th Natl. Conf. Saudi Comput. Colleges, NCCC 2022*, 2022.
- [69] H. Alamri, “Emotion Recognition in Arabic Speech From Saudi Dialect Corpus Using Machine Learning and Deep Learning Algorithms,” 2023.
- [70] M. E. M. Abo, N. Idris, R. Mahmud, A. Qazi, I. A. T. Hashem, J. Z. Maitama, U. Naseem, S. K. Khan, and S. Yang, “A multi-criteria approach for Arabic dialect sentiment analysis for online reviews: Exploiting optimal machine learning algorithm selection,” *Sustainability (Switzerland)*, vol. 13, 2021. doi: 10.3390/su131810018.
- [71] S. Alhumoud, T. Albuhairi, and W. Alohaideb, “Hybrid sentiment analyser for Arabic tweets using R,” in *IC3K 2015 - Proc. 7th Int. Joint Conf. Knowl. Discov., Knowl. Eng., Knowl. Manage.*, 2015.
- [72] G. Alwakid, “Sentiment analysis of dialectical Arabic social media content using a hybrid linguistic-machine learning approach,” Nottingham Trent University (United Kingdom), 2020.

- [73] W. M. A. Altohami and A. Omar, "A Computational-Augmented Critical Discourse Analysis of Tweets on the Saudi General Entertainment Authority Activities," *World J. Engl. Lang.*, vol. 12, 2022. doi: 10.5430/wjel.v12n8p471.
- [74] M. El-Masri, N. Altrabsheh, H. Mansour, and A. Ramsay, "A web-based tool for Arabic sentiment analysis," in *Procedia Computer Science*, 2017. doi: 10.1016/j.procs.2017.10.092.
- [75] S. and T. C. The Saudi Communications, "2023 Report of the Saudi Internet," 2023. Accessed: Jul.] 06, 2024. [Online]. Available: <https://www.cst.gov.sa/en/mediacenter/pressreleases/Pages/2024042402.aspx>
- [76] A. Radford, K. Narasimhan, and I. Sutskever, "GPT: Improving Language Understanding by Generative Pre-Training," *OpenAI*, 2018.
- [77] W. Antoun, F. Baly, and H. Hajj, "ARAGPT2: Pre-Trained Transformer for Arabic Language Generation," in *WANLP 2021 - 6th Arabic Natural Language Processing Workshop, Proceedings of the Workshop*, 2021.
- [78] S. Raschka, *Build a Large Language Model (From Scratch)*. in *From Scratch*. Manning, 2024. [Online]. Available: <https://books.google.com/books?id=sqG00AEACAAJ>
- [79] K. Papineni, S. Roukos, T. Ward, and W. J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2002.
- [80] A. Adadi and M. Berrada, "Peeking Inside the Black-Box: A Survey on Explainable Artificial Intelligence (XAI)," *IEEE Access*, vol. 6, 2018, doi: 10.1109/ACCESS.2018.2870052.
- [81] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should i trust you?" Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
- [82] A. El Mekki, A. El Mahdaouy, I. Berrada, and A. Khoumsi, "AdaSL: An unsupervised domain adaptation framework for Arabic multi-dialectal sequence labeling," *Inf. Process. Manag.*, vol. 59, 2022. doi: 10.1016/j.ipm.2022.102964.
- [83] A. Al-Essa, *When Najd meets Hijaz: Dialect contact in Jeddah*, 2009.

- [84] B. Flaij Alharbi and S. BinMasad, "A critical analysis of Saudi legal terms and their English translations," *Arab World English Journal for Translation and Literary Studies*, vol. 7, 2023. doi: 10.24093/awejtls/vol7no2.9.
- [85] A. Omar and B. D. Aldawsari, "The impact of diglossia on the language development and educational achievement of Saudi students in primary schools," *World Journal of English Language*, vol. 12, 2022. doi: 10.5430/wjel.v12n5p419.
- [86] Y. Aseri, K. Alreemy, S. Alelyani, and M. Mohanna, "Meeting challenges of modern standard Arabic and Saudi dialect identification," presented at the 2022.
- [87] N. Al-Twairish, H. Al-Khalifa, and A. Al-Salman, "Towards analyzing Saudi tweets," in *Proc. 1st Int. Conf. Arabic Computational Linguistics: Advances in Arabic Computational Linguistics*, ACLing 2015, 2016.
- [88] W. Al-Ghaith, "Developing lexicon-based algorithms and sentiment lexicon for sentiment analysis of Saudi dialect tweets," *Int. J. Adv. Comput. Sci. Appl.*, vol. 10, 2019.
- [89] M. D. Ahmed, "The emergence of a globalized city through multilingual local practice: A situated reading of signs in Jeddah," *J. Lang. Teach. Res.*, vol. 15, 2024. doi: 10.17507/jltr.1501.18.
- [90] D. N. Mati, M. Hamiti, A. Susuri, B. Selimi, and J. Ajdari, "Building dictionaries for low resource languages: Challenges of unsupervised learning," *Ann. Emerg. Technol. Comput.*, vol. 5, 2021. doi: 10.33166/AETiC.2021.03.005.
- [91] K. Abu Kwaik, M. Saad, S. Chatzikyriakidis, and S. Dobnik, "LSTM-CNN deep learning model for sentiment analysis of dialectal Arabic," in *Communications in Computer and Information Science*, 2019.
- [92] H. Elgibreen, M. Faisal, M. Sulaiman Al, S. Abdou, M. A. Mekhtiche, A. M. Moussa, Y. A. Alohal, W. Abdul, G. Muhammad, M. Rashwan, and M. Algabri, "An incremental approach to corpus design and construction: Application to a large contemporary Saudi corpus," *IEEE Access*, vol. 9, 2021. doi: 10.1109/ACCESS.2021.3089924.